WORKSHOP

Requested by the INTA committee

European Parliament

EU's trade and digital economy – Challenges and opportunities for small and medium enterprises (SMEs)



Authors:

Zaker AHMAD and Peter-Tobias STOLL

European Parliament coordinator:

External Policies Analysis and Support Unit

Directorate-General for External Policies of the Union

EN

Table of contents

EU's trade and digital economy – Challenges and opportunities for small and medium enterprises (SMEs)

How to strengthen innovation and intellectual property: Examples of Swedish SMEs and unicorns

DIRECTORATE-GENERAL FOR EXTERNAL POLICIES
EXTERNAL POLICIES ANALYSIS AND SUPPORT

European Parliament

IN-DEPTH ANALYSIS

EU's trade and digital economy – Challenges and opportunities for small and medium enterprises (SMEs)

ABSTRACT

This research paper examines the evolving landscape of digital trade and its impact on small and medium-sized enterprises (SMEs) in the European Union (EU). As digitalisation transforms global commerce, EU SMEs face significant opportunities as well as regulatory and competitive challenges. The study highlights key digital policy approaches, and in that context situates the current EU practice and commitments in international trade agreements affecting cross-border data flows, digital platforms and market access. The research highlights that while digital trade lowers costs and expands opportunities, complex compliance requirements and regulatory fragmentation hinder SME competitiveness. The study contrasts the EU's rights-based approach to digital governance with the market-driven framework of the United States and the state-controlled model of China. To enhance SME participation in digital trade, the paper recommends a coordinated whole-of-government approach to digital regulation, stronger EU leadership in global trade negotiations, tailored SME provisions in trade agreements and expanded support for digital innovation and skills.

EP/EXPO/2023/OP/0001/INTA/LOT6/1/C/4-1

EN

June 2025 – PE 754.479

© European Union, 2025

AUTHORS

- Zaker AHMAD, Georg-August University, Göttingen;
- Peter-Tobias STOLL, Georg-August University, Göttingen;

RESEARCH ASSISTANCE

- Jan THOMSEN, Georg-August University, Göttingen

PROJECT COORDINATOR (CONTRACTOR)

- Universität Innsbruck (PARLATRADE Consortium)

This paper was requested by the European Parliament's Committee on International Trade (INTA).

The content of this document is the sole responsibility of the authors, and any opinions expressed herein do not necessarily represent the official position of the European Parliament.

CONTACTS IN THE EUROPEAN PARLIAMENT

Coordination: Rasma KASKINA, External Policies Analysis and Support Unit

Editorial assistant: Balázs REISS

To give feedback or obtain copies, please write to exas@europarl.europa.eu.

VERSION

English-language manuscript completed in June 2025.

COPYRIGHT

Brussels © European Union, 2025

Reproduction and translation for non-commercial purposes are authorised, provided the source is acknowledged and the European Parliament is given prior notice and sent a copy.

© Image on cover page used under licence from Adobe Stock.

This paper will be published on the European Parliament's online database, 'Think Tank'.

Table of contents

1 Introduction 5

2 Digital trade and SMEs: EU state-of-play in the global context 6

2.1 The intersection of trade and the digital economy 6

2.1.1 Digital transformation and digital trade policy 6

2.1.2 Digital trade: regulatory dimensions and foreign practice 8

2.1.3 EU policy framework related to digital trade 10

2.1.4 Digital trade regulation and the WTO 13

2.2 Data protection and cross-border data flows: Impact on SMEs 14

2.2.1 The Landscape of data flow regulation in preferential agreements 14

2.2.2 SME interests in data flow regulation 17

2.3 The EU's Digital Services Act (DSA) and Digital Markets Act (DMA) 18

2.4 Digital trade regulation beyond data flows: Enabling and securing electronic commerce 21

2.5 Opportunities for EU SMEs in the global digital economy 23

3 Conclusions and recommendations 26

Bibliography 29

List of figures

Figure 1. Proportion of employment attributable to MSMEs in each industrial ecosystem (2023)... 8

Figure 2. Proportion of value-added by MSMEs in each industrial ecosystem (2023)..... 8

Figure 3. Degree of EU strategic autonomy in key technologies 12

1 Introduction

Simply put, the digital economy consists of new as well as traditional economic activities (i.e. production, movement, exchange and consumption of goods and services) that are enabled or influenced by the quantitative and qualitative improvements in digital technologies. According to the OECD, the information and communication technology (ICT) sector, which is the driver of the digital economy, grew three times faster during the 2013-2023 period compared to the rest of the OECD economy[1]. UNCTAD reports that between 2016 and 2022, business sales using e-commerce grew by 60 % – reaching a total of USD 27 trillion[2]. Digital trade is an integral part of this growth narrative. It refers to commerce enabled by digital means – by telecommunications and/or ICT services – and covers trade in both goods and services across all sectors of the economy.

It is therefore the foremost priority of government policy to ensure that businesses and communities can reap the full benefits of the digital economy while addressing the associated challenges and risks. The same applies to policies for micro, small and medium enterprises (SMEs)[3]. These enterprises are the lifeblood of the EU economy, both in terms of value added and employment. According to the 2024 Report on European SMEs, 99 % of all businesses in the non-financial business sector (NBFS) in the EU are SMEs, employing altogether 88.7 million people – almost 66 % of all NBFS employment in the EU[4]. SMEs account for more than 53 % of the annual NBFS value-added in the EU[5]. Many SMEs are directly engaged in delivering the horizontal, backbone services (e.g. computer programming) on which the digital economy is built. Others benefit from the digital infrastructure and platforms to access new demand, gain new knowledge and reduce business costs. As a result, on one hand, rules ensuring fairness and a level playing field in the digital space will help SMEs better compete with the larger business entities. On the other hand, new and detailed regulations that control the way data is transferred and used in the digital economy can add to the compliance costs for SMEs.

So far, states with significant stakeholdership and interests in the global digital market have adopted various approaches to digital regulation. The EU's rights-oriented regulation occupies a middle ground between the market-driven US approach and the state-centric Chinese approach. While EU regulations are at the forefront of establishing global standards for data and privacy protection, as well as regulating digital platforms and overseeing frontier technologies such as artificial intelligence, it has struggled to address its strategic dependencies in many of these areas. Similarly, although trade policies at various levels of governance (multilateral, plurilateral,

[1] OECD, *OECD Digital Economy Outlook 2024 (Volume 1): Embracing the Technology Frontier,* (2024), ch. 1, https://www.oecd-ilibrary.org/science-and-technology/oecd-digital-economy-outlook-2024-volume-1_a1689dc5-en.
[2] UNCTAD, *Business e-commerce sales and the role of online platforms*, UNCTAD Technical Notes on ICT for Development, No. 1, (2024), pp. 3-4, https://unctad.org/system/files/official-document/dtlecde2024d3_en.pdf.
[3] SMEs are officially defined as enterprises that have less than 250 employees – among which micro enterprises employ less than 10 and small enterprises employ less than 50 staff. *Commission Recommendation of 6 May 2003 concerning the definition of micro, small and medium-sized enterprises (Text with EEA relevance) (notified under document number C(2003) 1422)*, OJ L 124, (2003), Annex 1, http://data.europa.eu/eli/reco/2003/361/oj.
[4] European Commission, *Annual Report on European SMEs 2023/2024: SME Performance Review 2023/2024,* (2024), p. 6, https://data.europa.eu/doi/10.2826/355464.
[5] Annual Report on European SMEs 2023/2024, (n 4) above, p. 8.

preferential, and individual) can positively contribute to formally internationalising EU standards in the digital domain, the region is not currently the most active player in this regard.

The aim of this briefing is to provide a comprehensive overview of the aforementioned digital regulation and trade policy landscape, with a particular emphasis on its impact on SMEs. To achieve this aim, the following overview (Section 2) outlines the current knowledge and research on the intersection of trade and the digital economy, data protection, cross-border data flows and their effects on SMEs, the EU Digital Services Act and Digital Markets Act and their implications for international trade, as well as the opportunities for EU SMEs within the global digital economy. This is then followed by conclusions and recommendations (Section 3).

2 Digital trade and SMEs: EU state-of-play in the global context

2.1 The intersection of trade and the digital economy

2.1.1 Digital transformation and digital trade policy

The interest and concern over regulating digital trade occur within the broader context of accelerated digital transformation globally, particularly since the COVID-19-induced disruption of in-person social interactions. Digital transformation denotes the paradigm shift in the modes of action in all walks of life (e.g. social connectivity, education, healthcare, finance, work and business models, climate action) induced by greater and more innovative use of data and digital technologies (e.g. online marketplace, artificial intelligence (AI), cloud computing, internet of things (IOT)). In this sense, the notion 'digital transformation' is broader than competing concepts such as Industry 4.0 (also, 4th Industrial Revolution), which are used to only highlight manufacturing breakthroughs using intelligent computers[6]. Tapping into the benefits of digital transformation can expose societies to various forms of risks and inequalities such as issues involving privacy, safety and security, equality and protection of personal rights. To ensure a sustainable and inclusive digital transition, the OECD recommends a whole-of-government approach and cross-cutting policy actions ensuring access to digital infrastructure, balanced regulation of cross-border data flows, harnessing new technologies (e.g. AI and quantum technologies) in a safe manner and improving digital security[7].

The notion of 'digital trade' captures the transboundary and commercial dimensions of this significant shift. In its broadest sense, i.e. digitally enabled trade in goods and services, digital trade is not new. Computers and the Internet have been around for decades, and for example, a plurilateral agreement on removing customs duties on informational technology-related products

[6] McKinsey, *What are Industry 4.0, the Fourth Industrial Revolution, and 4IR?*, https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-are-industry-4-0-the-fourth-industrial-revolution-and-4ir

[7] OECD, *Digital Transformation*, https://www.oecd.org/en/topics/digital-transformation.html.

has been in place since 1996[8], and a work programme on e-commerce has existed since 1998[9]. However, with the exponential improvements in the ability to collect, store and compute large amounts of data, as well as advances made in algorithmic problem-solving techniques, what 'digitally-enabled' means has fundamentally changed over the past decade. To understand what digital trade comprises in the present-day world, three factors are worth noting. First and foremost, digital trade encompasses the exchange of traditional goods and services facilitated by digital means. Second, digital trade also encompasses trade in new digital-embedded items (e.g., a smartphone or a smart TV) or exclusively digital services (e.g., online platforms and marketplaces). Lastly, digital trade necessitates coming to terms with the ubiquity of data flows that underlie all forms of transactions[10].

For businesses, digital connectivity provides a 'double dividend' – by lowering trade costs on the one hand and increasing trade flows on the other. These effects are significant in both domestic and international trade. Gonzalez and others have found that increasing digital connectivity by 1 % leads to a 0.1 % lowering of trade costs and a 1.5 % increase in trade flows internationally[11]. The OECD notes that digitalisation is particularly favourable to micro, small and medium enterprises (MSMEs) as it enables cost-effective internationalisation potential by providing export opportunities and access to critical knowledge and information[12]. Digital platforms provide MSMEs easy access to facilitation services such as warehousing, e-payment and credit.

SMEs are present across all 14 industrial ecosystems[13], many of which are profoundly connected to or benefit from the digital economy. The top 3 ecosystems with the highest SME concentrations are retail (23 %), tourism (15 %), and cultural and creative industries (8.1 %)[14]. These industries are among the most significant users of digital technologies. SMEs are also active in the digital ecosystem (5.7 %), which encompasses services such as computer programming and information services. These form the essential backbone of the digital economy. In all these segments, SMEs

[8] WTO, *Ministerial Declaration on Trade in Information Technology Products*, WT/MIN(96)/16, (1996), https://www.wto.org/english/thewto_e/minist_e/min96_e/16.pdf. Scope of the agreement was expanded in 2015. See WTO, *Ministerial Declaration on the Expansion of Trade in Information Technology Products*, WT/MIN(15)/25, (2015), https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/MIN15/25.pdf&Open=True.

[9] See Section 2.1.4 below.

[10] These aspects are also highlighted in the OECD typology for digital trade. Javier López-González, Marie-Agnès Jouanjean, *Digital Trade: Developing a Framework for Analysis*, OECD Trade Policy Papers 205, (2017), https://www.oecd.org/en/publications/digital-trade_524c8c83-en.html.

[11] Javier López González, Silvia Sorescu, Pinar Kaynak, *Of bytes and trade: Quantifying the impact of digitalisation on trade*, OECD Trade Policy Papers, No. 273, (2023), pp. 17-22. https://doi.org/10.1787/11889f2a-en.

[12] OECD, *Key Issues in Digital Trade: OECD Global Forum on Trade 2023 "Making Digital Trade Work for All",* (2023), https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/01/key-issues-in-digital-trade-review_5a91f9a0/b2a9c4b1-en.pdf.

[13] The industrial ecosystem is a notion introduced by the European Commission to represent the EU industrial landscape in the form of a number of competence areas with strong inter-industry dependencies. The 14 identified ecosystems are: Aerospace and Defence, Agri-food, Construction, Cultural and Creative Industries, Digital, Electronics, Energy and Renewables, Energy Intensive Industries, Health, Mobility and Transport, Proximity and Social Economy, Retail, Textiles and Tourism. For further clarification, see Commission Staff Working Document: Annual Single Market Report 2021, SWD(2021) 351 final, European Commission, (2021), pp. 208–211, https://commission.europa.eu/system/files/2021-05/swd-annual-single-market-report-2021_en.pdf.

[14] Annual Report on European SMEs 2023/2024, (n 4) above, p. 12.

play a significant positive role, both in terms of employment creation and in terms of value-added. This is shown in Figures 1 and 2 below.

Figure 1. Proportion of employment attributable to MSMEs in each industrial ecosystem (2023)

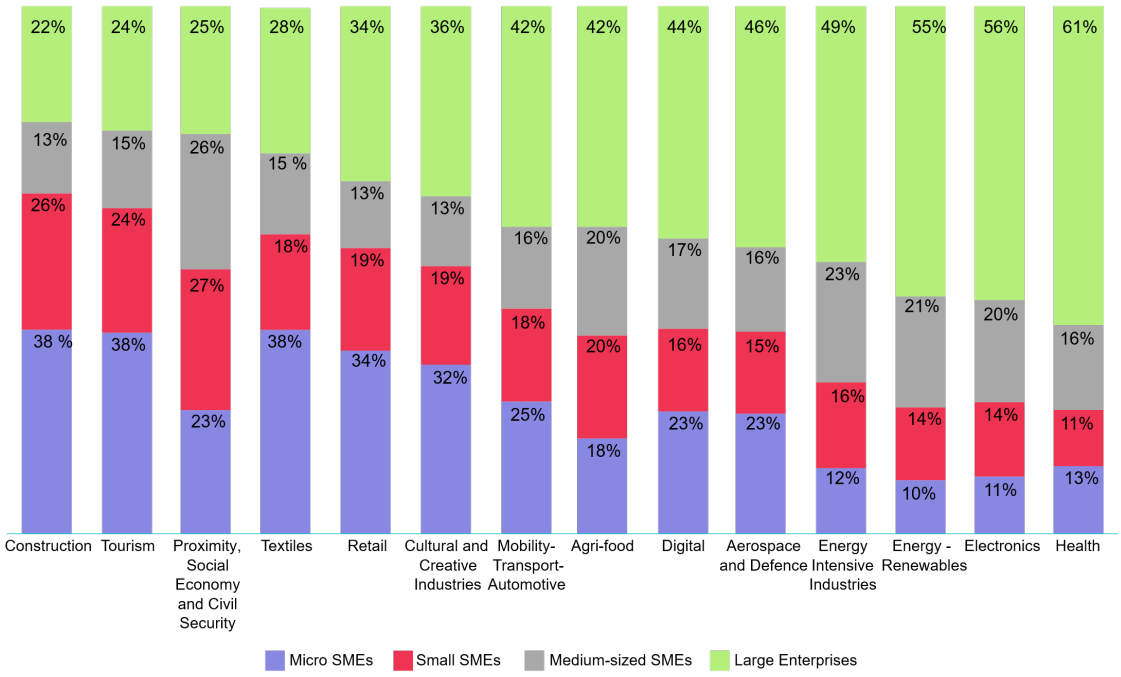
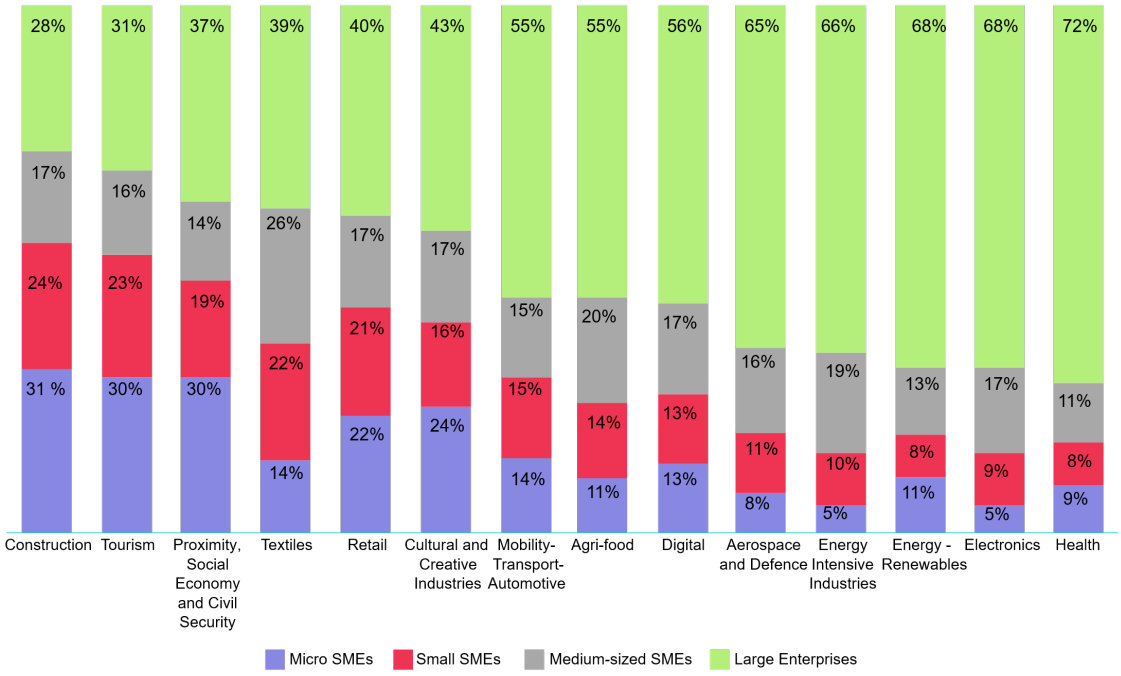


Figure 2. Proportion of value-added by MSMEs in each industrial ecosystem (2023)



Source: Annual Report on European SMEs (2024)

2.1.2 Digital trade: Regulatory dimensions and foreign practice

From a trade policy perspective, the digital economy presents various regulatory dimensions: (i) removal of digital trade barriers and facilitation measures, (ii) steps to ensure a level playing field

in the digital domain, (iii) incentivising innovation, (iv) ensuring safety and security, and (v) alignment with community values and human rights[15]. At present, policymakers' responses to these dimensions are distinct from one another in key digital markets rival to the EU – namely the United States of America (USA) and China. Key features of these markets are outlined below.

USA – The US is home to the most technology giants shaping the global digital revolution. US policy on digital regulation is dominated by a focus on free market principles, coupled with the preservation of internet freedom of speech. The US position is generally based on the understanding that cyberspace cannot be effectively regulated. Moreover, a free internet enhances democracy and is an essential aspect of free speech. Such weak oversight is also expected to leave the firms operating in this space the necessary freedom to operate, self-regulate and innovate. For example, Section 230 of the Communications Decency Act of 1996 in the US grants immunity to online platforms against being held responsible for the contents of third parties hosted by them[16]. US Federal Courts have found that such a step preserves the internet freedom and allows e-commerce to prosper[17]. The free-market approach also translates to a less interventionist approach or lack of regulation in other areas. US antitrust agencies often do not regulate the monopolistic behaviours of tech companies[18]. Also, there is no Federal law in the US protecting citizens' private data[19]. Despite the market favouring a regulatory model, the US government is active in combating online piracy and cybercrimes. However, the government itself also engages in extensive surveillance of online activities for national security purposes.

China – Diametrically opposite to the US market-driven regulatory approach is the Chinese command-and-control attitude. Key hallmarks of the Chinese policy approach include large-scale, opaque state support for technology firms, protection of the domestic market from foreign competitors, the absence of internet freedom, and very high levels of state surveillance. This combination has proven to be economically profitable for Chinese tech companies, such as Baidu, Alibaba, Tencent, or ByteDance – many of which now rival or surpass their US counterparts. The Made in China 2025 program, launched in 2015, has put government's industrial policy focus on

[15] A similar categorisation of digital regulation is done by the OECD. See OECD, *Digital Trade Inventory: Rules, Standards and Principles*, (2021), pp. 6–8, https://www.oecd-ilibrary.org/trade/digital-trade-inventory_9a9821e0-en.

[16] Protection for private blocking and screening of offensive material, 47 U.S.C. 230, https://www.govinfo.gov/app/details/USCODE-2011-title47/USCODE-2011-title47-chap5-subchapII-partI-sec230/summary. Section 230(c)(1) holds that "No provider or user of an interactive computer service shall be treated as the publisher or speaker of any information provided by another information content provider." For additional information, see Valerie C Brannon, Eric N Holmes, *Section 230: An Overview*, Congressional Research Service Report R46751, (2024), https://www.congress.gov/crs-product/R46751.

[17] Anu Bradford, *Digital Empires: The Global Battle to Regulate Technology* (Oxford University Press 2023), p. 46 et seq.

[18] This position was starting to shift under the Biden administration. For example, in a case brought the US Department of Justice (DOJ), the District Court of Columbia decided in 2024 that Google illegally monopolises the search engine market. See, 'United States v. Google, LLC', *Harvard Law Review*, vol. 138, (2025), p. 89, https://harvardlawreview.org/print/vol-138/united-states-v-google-llc/. With the new Trump administration, the approach may shift again.

[19] Some privacy laws exist at the state level. For example, California Consumer Privacy Act (CCPA) of 2018, https://oag.ca.gov/privacy/ccpa.

digital technologies including advanced information technology, automation, artificial intelligence and quantum computing[20]. For foreign technology firms, access to the Chinese markets is often conditioned on mandatory joint-venture agreements, data sharing and technology transfer commitments. Furthermore, China regulates the internet traffic in and out of the country in its entirety (a.k.a. the great firewall) – resulting in the blocking-at-will of digital services (e.g. YouTube or Google) supplied across borders. Heavy censorship and state surveillance of individual as well as business activities in China is already well-documented[21]. Chinese Cyber Security Law (CSL) requires critical infrastructure operators to store operational data within the territory of mainland China[22]. In recent years, the Chinese government has acted to protect the data and privacy of individuals[23] – only against other firms and not the government itself.

2.1.3 EU policy framework related to digital trade

The foremost defining feature of the EU's approach to digital regulation is its strong foundation in European values – 'respect for human dignity, freedom, democracy, equality, the rule of law and respect for human rights'[24]. This position is eloquently made in the European Declaration on Digital Rights and Principles for the Digital Decade[25], holding that technological solutions should, among others, respect human rights and enable solidarity and inclusion. In this backdrop, the digital regulatory landscape in the EU is framed by the following instruments.

The General Data Protection Regulation (GDPR) was enacted in 2016[26], laying down general requirements and standards applicable to the collection of personal data of individuals within the EU. Among other things, the GDPR requires obtaining explicit and specific consent from data subjects, lays down specific purpose and storage limitations, and imposes requirements for data integrity, confidentiality, and security. Non-compliance with the GDPR can result in very high fines. Since its introduction, the GDPR has become the global gold standard for data protection.

[20] The policy goal is inspired by the German *Industrie 4.0* policy. However, the implementation is very different. See, GIZ, *Made in China 2025 Factsheet*, (2018), https://www.plattform-i40.de/IP/Redaktion/EN/Downloads/Publikation/China/MIC2025_factsheet.pdf?__blob=publicationFile&v=3.

[21] Bradford, (n 17) above, pp. 77–91.

[22] *Translation: Cybersecurity Law of the People's Republic of China (Effective June 1, 2017)*, DigiChina, (2018), art. 37, https://digichina.stanford.edu/work/translation-cybersecurity-law-of-the-peoples-republic-of-china-effective-june-1-2017/. Article 31 elaborates critical infrastructures as those which, if affected, might seriously endanger national security, national welfare, the people's livelihood, or the public interest.

[23] *Personal Information Protection Law of the People's Republic of China*, (2021), https://personalinformationprotectionlaw.com/.

[24] Consolidated Versions of the Treaty on European Union and the Treaty on the functioning of the European Union, OJ C 202/1, (2016), art.2, https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:12016ME/TXT&from=EN.

[25] *European Declaration on Digital Rights and Principles for the Digital Decade*, 2023/C 23/01, (2023), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:JOC_2023_023_R_0001.

[26] Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation), OJ L 119, (2016), http://data.europa.eu/eli/reg/2016/679/oj (hereinafter 'GDPR').

The Digital Services Act (DSA) was enacted in 2022[27], laying down the standards of conduct for online platforms and intermediaries. Among others, the DSA requires the online platforms, especially the very large ones, to identify the systemic risks its hosted contents may pose, and address the same through effective, reasonable and proportionate measures. The platforms are also required to ensure that their algorithms operate in a fair, transparent and non-discriminatory manner. Non-compliance with the DSA is also followed up by hefty fines – up to 6 % of the platform's global turnover. More on the DSA and its impact appears later in this section.

The Digital Markets Act (DMA)[28] is an *ex-ante* tool in the hands of the Commission to ensure that digital markets operate in a fair and contestable[29] manner. The DMA identifies 'digital gatekeepers' – companies which have control over the access to information, services and opportunities in the digital world, and thereby can intermediate between a supplier and a consumer. Such gatekeepers should provide users with fair access, promote interoperability and refrain from discriminatory practices such as self-boosting/preferencing. More on the DMA and its impact is discussed later.

The Artificial Intelligence Act (AI Act)[30]**,** which came into force in 2024, is yet another first-ever attempt to mitigate the risks from AI (ab)use (e.g. manipulation, racial and social profiling, illegal biometric databases) on the one hand, and to encourage AI-related innovation in Europe. On the latter front, the regulation provides for the creation of AI 'regulatory sandboxes', providing a safe environment to train, test and validate AI models before placing on the market. SMEs shall have priority access to such sandboxes. Like the foregoing instruments, non-compliance with the AI Act incurs high penalties, ranging up to EUR 35 million (or 7 % of the global annual turnover) for engaging in prohibited AI practices and up to EUR 15 million (or 3 % of the global annual turnover) for high-risk activities.

The competitive landscape of digital policy and regulation is also informed by the EU's **Open Strategic Autonomy (OSA)** approach. The OSA highlights the strategic importance of achieving autonomy in key technology areas. In this regard, key digital infrastructural components such as cloud computing and artificial intelligence fall in areas where EU industry players currently fall behind. OSA, itself driven by geopolitical factors such as the technology rivalry with the USA and China, protectionism abroad and emerging tensions over national security, also influences the

[27] Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC (Digital Services Act), OJ L 277, (2022), http://data.europa.eu/eli/reg/2022/2065/oj (hereinafter, 'DSA').

[28] Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector and amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act), OJ L 265, (2022), http://data.europa.eu/eli/reg/2022/1925/oj (hereinafter, 'DMA').

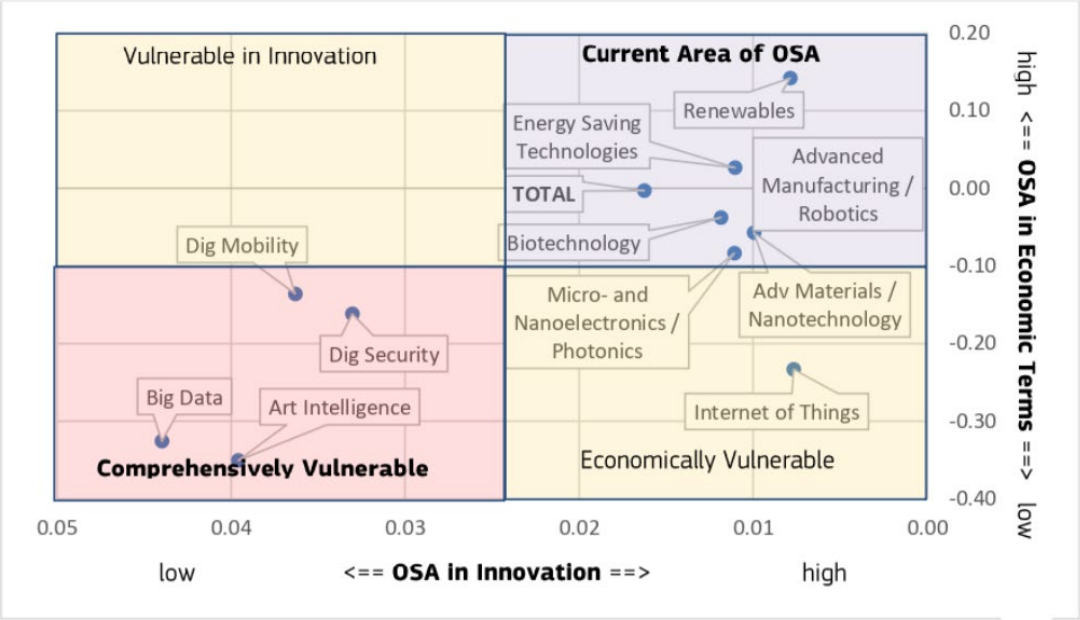
[29] In economic parlance, a market is contestable when there is very low barrier-to-entry. Due to this, even when a contestable market is served by one or few suppliers, they do not act in a monopolistic fashion. See Nigar Hashimzade, Gareth Myles, John Black, 'Contestable Market', *A Dictionary of Economics,* Oxford University Press, (2017), https://www.oxfordreference.com/display/10.1093/acref/9780198759430.001.0001/acref-9780198759430-e-557.

[30] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act), OJ L, 2024/1689, (2024), http://data.europa.eu/eli/reg/2024/1689/oj (hereinafter, 'AI Act').

evolution of digital regulation. Digital regulation is one of the two principal strands along which the green and digital transition is to take place in the coming decades. Through improved monitoring, management and use of energy using sensors, computer-aided analysis and artificial intelligence, digitalisation will play a crucial role in decarbonising the energy, construction, transport and agriculture sectors[31]. These reinforce the need to catch up on key strategic digital technology capabilities (e.g. blockchain, artificial intelligence and internet of things) and related infrastructure (e.g. 5G & 6G), as well as improve cybersecurity [32].

It is concerning that, particularly in the area of enabling digital technologies, the EU is in a position of strategic dependency. A 2024 research report examining OSA in terms of economic autonomy (level of reliance on imports)[33] and innovation autonomy (level of foreign reliance in domestic patenting activities)[34] finds that the digital technology sector is the worst-performing[35]. In particular areas such as AI and big data, net import reliance can be as high as 50 %. Figure 3 below shows the EU's comprehensive lack of strategic autonomy in key technologies.

Figure 3. Degree of EU strategic autonomy in key technologies



Source: Kroll (2024)

[31] European Commission, *2022 Strategic Foresight Report: Twinning the green and digital transitions in the new geopolitical context,* COM(2022) 289 final, (2022), pp. 4-8, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52022DC0289.

[32] COM(2022) 289 final, (n 31) above, pp. 13, 16.

[33] This is measured as the ratio between products available in the domestic market (including imports) and net import of that product.

[34] This is measured as the ratio between domestically registered patents and the number of foreign applications therein.

[35] Henning Kroll, Assessing Open Strategic Autonomy, Publications Office of the European Union, JRC136359, European Commission, (2024), pp. 21-23, https://publications.jrc.ec.europa.eu/repository/handle/JRC136359.

The above regulatory instruments and their underlying motivation also inform the Union's trade policy approach. The Commission's trade policy review in 2021 took special note of the increased servicification of trade, triggered as well as enabled by digital transformation, which is changing the nature of economic growth both internally and externally[36]. As a result, one of the mid-term goals set by the Commission is to support the digital transition and trade in services[37]. Specific tasks set out in this regard includes (i) continue to support WTO plurilateral negotiations on trade in services beyond e-commerce, (ii) establishing framework of cooperation on digital trade with third-country partners, and (ii) ensuring free flow of data in compliance with EU data protection rules and public policy goals[38]. The successful conclusion of WTO Joint Statement Initiative on e-commerce, and ongoing negotiations of digital trade agreements with key EU partners are examples of concrete steps taken to execute those tasks. Creation of a European analytical framework for measuring data flows was also planned[39]. The framework came to being in 2022[40], forming a key part of the European data strategy[41].

Based on the above hindsight, we conclude that the formulation of effective EU digital trade policy for the future should incorporate several key features. First, it should be ambitious to present a third, viable alternative to the market-driven US approach and the government-driven Chinese approach. In this connection, it would be important to ensure that the EU's trading partners, and key markets also share the same rights-based approach to data protection. Second, EU digital trade policy should continue to be flexible, to adapt with the fast-changing economic dynamics of new technologies. Third, to ensure coherence across regulatory instruments related to the digital economy that are taking shape under various EU regulatory bodies, a whole-of-government approach would be called for. Finally, appropriate care should be taken to ensure that compliance with the new regulation is not unreasonably burdensome for SMEs.

2.1.4 Digital trade regulation and the WTO

The WTO is a highly relevant platform to develop consensus among large groups of its members on issues related to the regulation of digital trade. A very early formation of the Work Programme on Electronic Commerce (1998)[42], is a testament to that fact. However, despite much discussion and exploration of the relationship between e-commerce and existing WTO rules, the program is

[36] European Commission, Communication from the Commission to the European Parliament, The Council, the European Economic and Social Committee and the Committee of the Regions – Trade Policy Review: An Open, Sustainable and Assertive Trade Policy, COM/2021/66 final, (2021), pp. 2–3.
[37] COM/2021/66 final, (n 36) above, p. 14.
[38] COM/2021/66 final, (n 36) above, p. 15.
[39] COM/2021/66 final, (n 36) above, p. 15.
[40] VVA and Tech4i2, *Study on Mapping Data Flows: Final Report*, European Commission, (2021), https://data.europa.eu/doi/10.2759/7355.
[41] European Commission, *European Data Strategy*, https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/european-data-strategy_en. Other aspects of the European data strategy includes facilitation of data access availability and sharing across Member States and sectors through European Data Governance Act (Reg. 2022/868, in force: 2024) and European Data Act (Reg. 2023/2854, in force: 2024).
[42] WTO, *Work Programme on Electronic Commerce*, WT/L/274, (1998), https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/L/274.pdf&Open=True.

yet to come up with any multilateral rules. A moratorium on imposition of customs duties on electronic transmissions was also introduced in 1998, which has continued until this date. However, it was decided at the 13th Ministerial Conference (MC13) of the WTO that the moratorium as well as the program itself is to come to an end by MC14 or by 31 March 2026, whichever comes earlier[43]. Although calls for making the moratorium permanent were made, it was not possible due to lack of finding a common ground with developing country interests[44].

Since 2017, like-minded WTO members are engaged in plurilateral negotiations – known as a Joint Statement Initiative (JSI) on the trade-related aspects of e-commerce. The group currently comprises 91 members. In 2024, the JSI released an Agreement of Electronic Commerce[45]. Key areas covered by the new agreement include e-commerce facilitation, moratorium on customs duties, data protection, consumer protection and cybersecurity. Existing general exception, security exception and dispute settlement provisions are extended to cover this agreement. To formally turn it into a WTO plurilateral agreement, the members need to present it at the MC14[46], to be included into the relevant annex (Annex 4) of the WTO Agreement. This decision can only be taken by consensus[47]. The key hurdle in this regard will be to secure supports from those WTO Members (e.g. India) who have persistently objected to plurilateral disciplines as well as e-commerce negotiations. Thereafter, the agreement must also obtain a 'critical mass'[48] of acceptance for it to enter into force.

2.2 Data protection and cross-border data flows: Impact on SMEs

2.2.1 The landscape of data flow regulation in preferential agreements

The divergent third-country practice on data flow regulation in key digital markets has already been outlined above. Coupled with the lack of progress at the WTO on digital trade issues, these different narratives currently compete for dominance in the regulatory space of free trade agreements. Expert research reveals that one out of every ten FTAs contains some provision on free movement of data[49]. Among these, 30 agreements contain legally binding commitments – 16 of which were signed between 2020 and 2023[50]. Out of those 16, only two include the EU as a

[43] WTO, *Ministerial Decision: Work Programme on Electronic Commerce*, WT/MIN(24)/38; WT/L/1193, (2024), https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/WT/MIN24/38.pdf&Open=True.
[44] Wallace Cheng, Clara Brandi, *Governing Digital Trade – a New Role for the WTO*, Briefing Paper 6/2019, (2019), https://www.idos-research.de/uploads/media/BP_6.2019.pdf.
[45] WTO, *Joint Statement Initiative on Electronic Commerce – Agreement on Electronic Commerce,* INF/ECOM/87, (2024), https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/INF/ECOM/87.pdf&Open=True#page=14.47 (hereinafter, 'E-Commerce JSI').
[46] MC14 is set to take place in Cameroon, from 26 to 29 March 2026.
[47] Marrakesh Agreement Establishing the World Trade Organization, 1867 UNTS 154; 33 ILM 1144, (1994), art. 10.9.
[48] E-Commerce JSI, (n 45) above, art. 29.2.
[49] Mira Burri, María Vásquez Callo-Müller and Kholofelo Kugler, 'The Evolution of Digital Trade Law: Insights from TAPED', *World Trade Review,* vol. 23, (2024), p. 196.
[50] Burri, Vásquez Callo-Müller and Kugler, (n 49) above, pp. 196–198.

party[51]. To those two, are now added two new EU digital trade agreements (DTA) with Singapore and Korea[52]. The EU DTAs are yet to be formally signed by the parties.

The practice of stringent digital trade and data flow commitments in preferential trade agreements was pioneered by the USA – first as the chapter on e-commerce in the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP)[53], later as the chapter on digital trade in the United States-Mexico-Canada Agreement (USMCA)[54]. However, references to e-commerce in trade agreements date back to the early 2000s[55]. Both CPTPP and USMCA require parties to allow free cross-border electronic transfer of information, subject to the adoption or maintenance of any measure necessary to pursue legitimate policy objectives[56].

Although relatively slow compared to selected other countries such as New Zealand, Singapore and Australia, the EU is gradually spreading its own approach to data flow and digital trade regulation[57] across borders by embedding it in trade agreements with partners. This has been termed by one expert as *de jure* internationalisation[58]. While committing to the international practice of free data flows, the EU approach conditions its implementation in unique ways. First, free flow of data and prohibition of data localisation are implemented subject to the general exceptions, and also 'sympathetic consideration' by parties of any necessary subsequent changes of policy[59]. Second,

[51] Trade and Cooperation Agreement between the European Union and the European Atomic Energy Community, of the One Part, and the United Kingdom of Great Britain and Northern Ireland, of the Other Part, OJ L 149, (2021), http://data.europa.eu/eli/agree_internation/2021/689(1)/oj/eng (hereinafter, 'EU-UK TCA'); Free Trade Agreement between the European Union and New Zealand, OJ L, 2024/866, (2024), http://data.europa.eu/eli/agree_internation/2024/866/oj (hereinafter, 'EU-New Zealand FTA'). The interim trade agreement (ITA) signed by EU with Chile, which entering into force in February 2015, also contains similar commitments on data flow regulation.

[52] Agreement on Digital Trade Between the European Union and the Republic of Singapore, https://circabc.europa.eu/ui/group/09242a36-a438-40fd-a7af-fe32e36cbd0e/library/66ccfa9f-e239-4893-8e12-64f8ff1d1221/details?download=true (hereinafter, 'EU-Singapore DTA'); Agreement on Digital Trade Between the European Union and the Republic of Korea, https://circabc.europa.eu/ui/group/09242a36-a438-40fd-a7af-fe32e36cbd0e/library/1bddb97a-c02e-41e6-95d1-6e41029c880f/details?download=true (hereinafter, 'EU-Korea DTA').

[53] 'Chapter 14: Electronic Commerce', *Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP)*, (2018), https://www.mfat.govt.nz/assets/Trade-agreements/TPP/Text-ENGLISH/14.-Electronic-Commerce-Chapter.pdf (hereinafter, 'CPTPP').

[54] 'Chapter 19: Digital Trade', *Agreement between the United States of America, the United Mexican States, and Canada*, (2020), https://ustr.gov/sites/default/files/files/agreements/FTA/USMCA/Text/19-Digital-Trade.pdf (hereinafter 'USMCA').

[55] José-Antonio Monteiro, Robert Teh, *Provisions on Electronic Commerce in Regional Trade Agreements*, WTO working paper, ERSD-2017-11, (2017); The first EU commitment on e-commerce appears in 2008, in the EU-CARIFORUM Economic Partnership Agreement (EPA). Marc Jütten, *Briefing: The EU's Digital Trade Policy*, EPRS, (2024), https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/757615/EPRS_BRI(2024)757615_EN.pdf.

[56] CPTPP, art. 14.11; USMCA, art. 19.11.

[57] The Commission highlights six core elements of EU approach to digital trade, which can be found here: European Commission, *Digital Trade in EU Trade Agreements*, https://trade.ec.europa.eu/access-to-markets/en/content/digital-trade-eu-trade-agreements-0.

[58] As opposed to *de facto* internationalisation of EU standards, taking place through their integration in the big tech corporations' global policies (e.g. privacy standards). Anu Bradford, *The Brussels Effect: How the European Union Rules the World,* Oxford University press, (2020), pp. 142–147.

[59] EU-UK TCA, art. 201; EU-New Zealand FTA, art. 12.4.

privacy is recognised as a fundamental right, calling for high standards of protection[60]. To that effect, the parties preserve the policy room to maintain appropriate safeguards (e.g. GDPR) to ensure protection of personal data and privacy. Additionally, the right to regulate on public policy grounds – including traditional areas such as health, safety, public morals and new ones such as social services, education, consumer protection, data protection and cultural diversity – is explicitly incorporated[61]. Furthermore, the EU also uniquely reserves the right to review the commitments in several years' time.

The EU DTAs with Singapore and Korea address data flow regulations in line with the above approach, but provide greater detail. The DTAs are supplements to existing preferential agreements with the respective trading partners. Parties reaffirm their commitment to free data flows and prohibit data localisation, subject to an implementation review in three years, as well as maintenance of exceptions necessary to pursue 'legitimate policy objectives'[62]. The latter term is explained to cover traditional (e.g. health, environment, public morals, etc.) and new (e.g. online safety, cybersecurity, trustworthy AI etc.) policy priorities, taking into account the evolutionary nature of digital technologies[63]. The DTAs elaborate the standards of personal data protection and privacy in greater detail. In addition to recognising individuals' right to privacy and data protection, the new commitments oblige each party to adopt legal instruments, in line with international guidelines and principles[64], to protect personal data in a non-discriminatory manner.

However, the scope of this commitment varies. In case of the EU-Korea DTA, the strict obligation to adopt a data protection law is only regarding "individuals engaged in digital trade"[65]. For other situations, Article 6(4) of the agreement incorporates a traditional reservation of the right of each party to adopt and maintain measures on personal data and privacy protection. In the case of the EU-Singapore DTA, the obligation to adopt data protection laws apply to all individuals, and not only those engaged in digital trade[66]. The traditional reservation clause protecting measures taken to ensure data protection remains present, nevertheless. Both agreements contain additional obligations to ensure transparency and guidance regarding compliance with the requirements as well as available remedies. Additionally, the EU-Singapore DTA expresses the desire for the parties' legal approaches to personal data protection and data-flow regulation to converge over time through mechanisms such as mutual recognition, joint guidelines or compliance with broader international frameworks[67].

[60] EU-UK TCA, art. 202; EU-New Zealand FTA, art. 12.5.
[61] Mira Burri, 'Digital Trade: New Design Elements in Preferential Trade Agreements', *The Concept Design of a Twenty-First Century Preferential Trade Agreement: Trends and Future Innovations,* Cambridge University Press, (2025), pp. 160–161.
[62] EU-Korea DTA, art. 5; EU-Singapore DTA, art. 5.
[63] EU-Korea DTA, fn. 5; EU-Singapore DTA, fn. 1.
[64] EU-Singapore DTA specifically refers to the OECD Guidelines Governing the Protection of Privacy and Trans-Border Flows of Personal Data.
[65] EU-Korea DTA, art. 6(2).
[66] EU-Singapore DTA, art. 6(2).
[67] EU-Singapore DTA, art. 6(7).

As we see above, data flow and data protection-related commitments in preferential agreements tend to be coupled with ample reservation of policy space to pursue legitimate objectives. In some instances, such policy space has served parties to continue to maintain their domestic laws on data localisation. For example, the Regional Comprehensive Economic Partnership Agreement (RCEP)[68], which has 15 members including Australia, New Zealand, Singapore, Japan, China and Vietnam, maintains broad discretionary clauses to adjust for its members' regulatory divergence on data protection. Like other agreements, RCEP requires free flow of data to conduct business (Article 12.15) and prohibits data localisation requirements as a precondition (Article 12.14) in that regard. Also, like other agreements, these commitments are subject to legitimate policy objectives. Uniquely, however, RCEP provides that the implementing party can decide the legitimacy of a policy objective solely on its own, thereby giving these exceptions an almost 'self-judging' status[69]. Additionally, it is recognised that interests such as security or confidentiality may motivate parties to adopt data localisation strategy. Furthermore, countries such as Vietnam are granted limited exemptions from compliance with these commitments[70].

2.2.2 SME interests in data flow regulation

SMEs are well-integrated into each industrial ecosystem in the EU. Hence, how data flow regulation impacts SMEs depends on the nature of the impact in each ecosystem. To the extent free flow of data and digital innovation improves the manner, efficiency and costs in the end-use industries (e.g. retail), the SMEs stand to benefit greatly. The 2024 report on European SMEs mentioned that SMEs tend to dominate in the user, distributor and installer segments of the value chain. Their role as users of technologies such as AI and cloud computing is crucial, as these sectors cannot mature in the EU without efficient users whose feedback will guide future development trajectories[71]. In particular, SMEs, as users of digital tools, would benefit from a level playing field in online platforms and marketplaces.

On the other hand, for SMEs active on the innovation side of the digital landscape, increased regulation can bring mixed results. One traditional argument, made most vocally by the current US government as well as its big technology firms[72], is that higher regulation adds to the costs of compliance, which deter digital entrepreneurs from entering the EU market. The same line of argument may also apply to SMEs – more so due to the latter's much smaller financial footprint. In

[68] Regional Comprehensive Economic Partnership Agreement, (2020) https://www.dfat.gov.au/trade/agreements/in-force/rcep/rcep-text (hereinafter, 'RCEP').
[69] RCEP, ch. 12 nn. 12, 14.
[70] Vietnam is granted 5 years of exemption from the date of RCEP's entry into force.
[71] Annual Report on European SMEs 2023/2024, (n 4) above, p. 54.
[72] Bloomberg, *Trump Administration Pressures Europe to Reject AI Rulebook*, (25 April 2025), https://www.bloomberg.com/news/articles/2025-04-25/trump-administration-pressures-europe-to-reject-ai-rulebook; POLITICO, *Trump's Antitrust Agency Chief Blasts EU Digital Rules as "Taxes on American Firms"*, (2 April 2025), https://www.politico.eu/article/trumps-antitrust-agency-chief-blasts-eu-digital-rules-as-taxes-on-american-firms/; See for example, POLITICO, *Zuckerberg: It's "Sad" That EU Is Left behind on AI*, (19 December 2024), https://www.politico.eu/article/zuckerberg-its-sad-that-eu-is-left-behind-on-ai/.

2024, a key Commission report on the future of European competitiveness[73] also issued similar findings. The report identifies that, in addition to shortcomings in industrial dynamism, public research funding and the coherence of the EU single market, the regulatory burden in the EU is particularly difficult to bear for young tech companies[74]. According to a survey conducted by the European Investment Bank (EIB), 60 % of all businesses find regulations to be a challenge to making new investments, and approximately 25 % find regulations to be a significant obstacle[75]. However, improved regulation of competition in the digital domain by, among others, preventing the bigger players from abusing their control over the digital platforms makes it easier for smaller innovative firms to compete fairly. Bradford and others recommend more caution against drastic deregulation in the digital domain for it may cost EU its global position as the champion of rights[76]. Also, wholesale deregulation may not, by itself, be sufficient to trigger the necessary innovation and digital entrepreneurship.

Some of the existing preferential agreements, such as the DTAs negotiated by the EU, already recognise the necessity of ensuring that digital trade supports entrepreneurship, innovation, competition and inclusiveness to the benefit of SMEs. However, actual commitments do not yet go much beyond soft law obligations of cooperation and information sharing. For example, both EU-Singapore and EU-Korea DTAs hold that the "Parties shall endeavour to exchange information and best practices" regarding technologies that improve SMEs' digital trade opportunities[77]. The EU-Singapore DTA further adds a soft cooperation obligation regarding identifying ways to improve access to government data by SMEs with a view to generating new business and research opportunities[78].

2.3 The EU's Digital Services Act (DSA) and Digital Markets Act (DMA)

The DSA and the DMA represent landmark legislation for the specific regulatory challenges associated with digital platforms. While the DSA focuses on the regulation of online content, safety and the consumers' rights, the DMA aims to prevent anti-competitive conduct originating from the dominant market positions of the designated gatekeepers, promoting a level playing field for businesses. Therefore, both regulations strive to implement European core values of transparency, consumer protection and fair competition in the digital economic space.

[73] European Commission, The future of European Competitiveness, (2024), https://commission.europa.eu/document/download/97e481fd-2dc3-412d-be4c-f152a8232961_en#page=3.00.
[74] The future of European Competitiveness, (n 73), pp. 28-30.
[75] European Investment Bank, *Investment Barriers in the European Union 2023: A Report by the European Investment Bank Group,* (2024), pp. 32–33, 54, https://data.europa.eu/doi/10.2867/364809. The report also identifies that US firms are more likely to identify business regulation as a barrier than the EU firms.
[76] Anu Bradford, R Daniel Kelemen and Tommaso Pavone, 'Europe Could Lose What Makes It Great', *Foreign Affairs,* (21 April 2025), https://www.foreignaffairs.com/europe/europe-could-lose-what-makes-it-great.
[77] EU-Korea DTA, art. 23; EU-Singapore DTA, art. 24.
[78] EU-Singapore DTA, art. 16.

The regulations are likely to affect international trade in a multitude of ways. Most notably, the compliance costs could act as digital trade barriers as well as market entry barriers. Non-EU companies may not be able to transfer their business models from other jurisdictions into the EU digital market. This especially affects companies whose business models are centred around data collection, personalised advertising or algorithms which under EU law would unlawfully disadvantage users or competitors. As a result, such undertakings may be deterred from entering the EU digital market altogether.

The EU enforces the provisions through its new ability to impose substantial fines on companies in violation of the regulations[79]. The European Commission has already used this mechanism under the DMA for the first time, when it imposed fines on Apple (EUR 500 million) and Meta (EUR 200 million)[80]. The Commission found Apple to be non-compliant with obligations under the DMA, not allowing app developers and distributors to freely inform consumers of alternatives outside of App Store and allowing the consumers to make their purchases there. Meta, on the other hand, was not compliant with provisions prohibiting gatekeepers from combining users' personal data between services without their consent.

For the DSA, while there have been no final decisions on the imposition of fines, the social media company X (formerly Twitter) has been presented with preliminary findings, concerning possible violations of the DSA with respect to dark pattern, ad transparency and data access[81]. If a fine is imposed, it could be as high as up to 6 % of the annual global turnover of X. Furthermore, the DSA economically impacts tech corporations through the obligation to pay supervisory fees. These fees, which may amount up to 0.05 % of a corporation's global annual turnover, are raised to fund the costs the supervision. Currently there are several pending court cases dealing with legitimacy and proportionality of the fees[82]. At the same time, several formal proceedings concerning possible infringements of the DSA are underway[83]. These fines and fees demonstrate the financial risks

[79] See DMA, art. 30; DSA, art. 52.

[80] European Commission, *Press Release – Commission finds Apple and Meta in breach of the Digital Markets Act,* (23 April 2025), https://ec.europa.eu/commission/presscorner/detail/en/ip_25_1085.

[81] European Commission, *Press Release – Commission Sends Preliminary Findings to X for Breach of DSA,* (12 July 2024), https://ec.europa.eu/commission/presscorner/detail/en/ip_24_3761.

[82] *Application – Meta Platforms Ireland v Commission,* Case T-55/24, Court of Justice of the European Union (CURIA), (2024), https://curia.europa.eu/juris/document/document.jsf?text=2022%252F2065&docid=284221&pageIndex=0&doclang=en&mode=lst&dir=&occ=first&part=1&cid=4724475#ctx1; *Application – Tiktok Technology v Commission,* Case T-58/24, Court of Justice of the European Union (CURIA), (2024), https://curia.europa.eu/juris/document/document.jsf?text=supervisory%2Bfee%252C%2Bdigital%2Bservices%2Bact&docid=284229&pageIndex=0&doclang=EN&mode=req&dir=&occ=first&part=1&cid=2226762#ctx1.

[83] For example proceedings against Facebook and Instagram, see European Commission, *Press Release – Commission opens formal proceedings against Facebook and Instagram under the Digital Services Act*, (30 April 2024), https://ec.europa.eu/commission/presscorner/detail/en/ip_24_2373. The Regional Court (*Landgericht*) of Berlin has issued an injunction, ordering X to provide plaintiffs data access under DSA Art. 40. See Library of Congress, *Germany: Berlin Court Orders Social Media Platform X to Share Public Data on Elections with Researchers*, https://www.loc.gov/item/global-legal-monitor/2025-02-20/germany-berlin-court-orders-social-media-platform-x-to-share-public-data-on-elections-with-researchers/.

global tech companies face when operating in the European market and illustrate the EU's commitment to enforcing the regulations, thus, amplifying their impact on international trade.

Nonetheless, the impact of these fines and fees is not limited to the economic dimension, but also entails political implications. As most of the designated gatekeepers[84] under the DMA and the companies encompassed by the DSA are based in the USA, the obligations under these regulations as well as the potential imposition of fines has been interpreted as an unwarranted targeting of their tech sector[85]. Therefore, the enforcement of the regulations could further strain the existing tensions in the trading relations between the EU and the USA.

Nevertheless, international digital trade also benefits from the increase in competitiveness of the EU digital markets facilitated particularly by the DMA. It can be expected that the regulations foster an increase of trade and investment flows by promoting a more competitive, transparent and overall fair environment for digital markets[86].

Furthermore, the regulations may influence the operations of the digital platforms, intermediaries and markets globally through the so-called "brussels effect" – a phenomenon where multinational corporations implement the stringent EU standards worldwide to ensure coherence[87]. Brussels effect is already observable for other innovative pieces of legislation adopted by the EU, especially the GDPR which has become the global benchmark for the regulation of personal data. Currently, it does not seem likely that the DSA or the DMA will prove to have such an effect. Big tech corporations have demonstrated the capability of adjusting their strategies, so as to operate effectively in different jurisdictions with differing levels of regulatory oversight in the digital sector[88]. Similarly, theoretically, the EU regulation of online platforms and digital marketplaces could serve as a model for other jurisdictions. However, so far there has been no observation of third-country legislation being modelled after the DSA or the DMA.

Although the primary addressees of the DSA and DMA are big tech corporations, the regulations also have considerable effects on SMEs. In prior efforts to regulate the digital economic space, concerns were voiced that these may prove to be disproportionately burdensome for SMEs due to the compliance costs (especially for the GDPR). An overregulation with the entailing compliance costs could hinder the innovation, growth and competitiveness of European SMEs. DMA and DSA have been formulated with such concerns in mind. As already explained, the DMA applies to 'gatekeepers' – a role in which an SME is highly unlikely to find itself. The DSA, which regulates

[84] As of May 2024, the designated gatekeepers are Alphabet, Amazon, Apple, Booking, ByteDance, Meta and Microsoft. See European Commission, *DMA Designated Gatekeepers*, (2024), https://digital-markets-act.ec.europa.eu/gatekeepers_en.

[85] Donald J Trump, *Memorandum on Defending American Companies and Innovators From Overseas Extortion and Unfair Fines and Penalties*, (2025), https://www.whitehouse.gov/presidential-actions/2025/02/defending-american-companies-and-innovators-from-overseas-extortion-and-unfair-fines-and-penalties/.

[86] European Commission, *DMA Impact Assessment Report*, SWD(2020) 363 final Part 1/2, (2020), https://digital-strategy.ec.europa.eu/en/library/impact-assessment-digital-markets-act.

[87] Bradford, (n 58) above, ch. 2.

[88] Martin Eifert, Axel Metzger, Heike Schweitzer and Gerhard Wagner, 'Taming the Giants: The DMA/DSA Package', *Common Market Law Review,* vol. 58, (2021), pp. 1027-1028, https://doi.org/10.54648/cola2021065.

standards of conduct for online services and platforms, is equally applicable on SMEs that either provide or use digital services. Whenever fixed regulatory costs apply equally on companies of different sizes, the smaller companies (SMEs, in this case) suffer disproportionately more. However, not only are obligations under the DSA are tiered, i.e. made proportionate to the role and influence of the digital service provider in the cyberspace, but there are also specific built-in exclusions for SMEs providing intermediary or online platform services[89]. These concerns were adequately addressed by ensuring that the scope of these regulations does not encompass SMEs, thereby, not being disproportionally burdensome. The Commission is also mandated to conduct a review of the DSA, which shall include evaluation of the impact of DSA obligations on SMEs[90].

The DMA benefits SMEs by enhancing their ability to compete in the digital market. As a result of the maintenance of a competitive business environment, the opportunities for SMEs to utilise the advantages of the digital single market grow. For example, the prohibition of self-preferencing can reduce the obstacles SMEs face when offerings apps or services over the gatekeepers' platform, enabling them to operate with greater visibility and, better access to users. Therefore, the DMA allows SMEs to compete in a fairer digital market by reducing anti-competitive conduct of the designated gatekeepers. From the perspective of SMEs this is especially welcome, when taking into account the importance of digital markets and digitalisation for their international expansion.

2.4 Digital trade regulation beyond data flows: Enabling and securing electronic commerce

In addition to ensuring free flow of data and high standards of data protection, digital trade regulation play an increasingly significant role in several other aspects of the global digital economy. An overview of the existing and emerging issues in this regard is provided below.

Regulation of electronic commerce is the foremost area where trade policy can play an instrumental and positive role. Two issues have been of longstanding importance: (i) the suspension of customs duties on electronic transmissions and digitally transferred content, and (ii) the non-discriminatory treatment of digital products. Achieving broad international consensus on these issues is fundamentally important for securing a predictable and cost-effective trading environment for the SMEs. In absence of multilateral progress on these accounts, preferential trade agreements such as the CPTPP[91] as well as new DTAs incorporate these commitments. EU's preferential agreements with the UK and New Zealand as well as DTAs with Korea and Singapore commit to refraining from imposing customs duties on e-commerce. However, there is no commitment on non-discriminatory treatment of digital products. The RCEP is silent on both these issues, although many of its members (e.g. Australia, Japan, New Zealand, South Korea and Vietnam) are parties to other preferential agreements that subscribe to such commitments. Among other developing countries, India and South Africa remain reluctant to support any general moratorium on e-commerce duties.

[89] DSA, arts 15(2), 19, 29.
[90] DSA, Preamble, para. 150.
[91] CPTPP, art. 14.3; See also USMCA, arts 19.3, 19.4.

Most notably, India's recent agreements with some of the forerunners of digital trade rules like Australia or the European Free Trade Association (EFTA) countries contain no provisions on digital trade[92]. Without the developing countries' support, the WTO JSI Agreement on Electronic Commerce will fail to be adopted as a plurilateral agreement.

Facilitation of e-commerce transactions through the development and adoption of shared rules on electronic signatures, contracts and payments, promotion of paperless trade and sharing of electronic information between single windows[93] are win-win solutions for reducing the overall cost of trade. These are beneficial for all businesses including the SMEs. A report by the United Nations recently noted that full digital trade facilitation, i.e. facilitating paperless trade in additional to full implementation of the WTO commitments under the Trade Facilitation Agreement, can lower global trade costs by 14 %[94]. on average. Existing preferential agreements display significant convergence on digital trade facilitation commitments. Burri notes that the number of preferential agreements that contain rules on electronic signatures, paperless trading and electronic transactions frameworks amount to a hundred, ninety-four and eighty-five respectively[95]. Digital trade facilitation is also part of the WTO JSI Agreement on Electronic Commerce[96].

For EU enterprises including SMEs active in the digital industrial ecosystem, it is important to ensure adequate protection of their proprietary source codes and algorithms. Requiring such entities to share their digital intellectual property (IP) in exchange of market access rights is unfair and harmful. For example, there is a concern that the Chinese Cybersecurity Law can be used by the government to demand foreign companies to share their proprietary source code[97]. Similarly, under China's proposed AI regulation, a firm must file its algorithms with the authority when its AI-based product or service has social mobilisation capabilities[98]. However, preferential agreements such as the CPTPP incorporates prohibitions on requirements of sharing the source codes of mass market software, as a precondition for import or sale. Protection of software source codes is also a core element of the EU approach to digital trade in preferential agreements[99]. However, in 2023, USA withdrew its support for data flow, data localisation and source code provisions in the context

[92] Kholofelo Kugler, The Great Disconnect in India's Digital Trade Policy, Council on Foreign Relations, (2025), https://www.cfr.org/article/great-disconnect-indias-digital-trade-policy.
[93] Single windows are digital platforms setup by governments to provide traders with one-stop solution with the procedural formalities. Protection, harmonisation and interoperability of data elements across single windows facilitate both traditional and e-commerce.
[94] United Nations, Digital and Sustainable Trade Facilitation: Global Report 2023, (2023), pp. 58–59, https://repository.unescap.org/server/api/core/bitstreams/fbffab6f-223b-447a-a6ff-8f4bea16d6e7/content#page=70.59. The rate varies across countries. While pacific island nations will benefit from an almost 26 % reduction in trade costs, for developed economies it will result in approximately 7 % reduction.
[95] Burri, (n 61) above, n 12.
[96] E-Commerce JSI, (n 45), arts 4-10.
[97] Cybersecurity Law of China, (n 22) above, art. 35.
[98] Latham & Watkins, *China's New AI Regulations,* (2023), pp. 5–6, https://www.lw.com/admin/upload/SiteAttachments/Chinas-New-AI-Regulations.pdf.
[99] European Commission, *Digital trade in EU trade agreements,* https://trade.ec.europa.eu/access-to-markets/en/content/digital-trade-eu-trade-agreements-0.

of the WTO JSI discussions on e-commerce[100]. As a result, the final agreement is devoid of any prohibition on requirements to share source codes and/or algorithms.

Regulation of digital trade also triggers security concerns along multiple dimensions. At the user-level, the interest to secure data and privacy has resulted in contrasting data protection and localisation standards, which have been elaborated above[101]. For digital companies, securing proprietary code and other digital IP calls for necessary protection agreed to that effect in preferential agreements. At the EU level, the Cyber Resilience Act[102] (CRA) aims to enhance cybersecurity across digital products by mandating security updates and establishing standards for product security. The CRA includes provisions tailored for SMEs, such as SME-specific guidance and support in implementing cybersecurity standards. Preferential agreements contribute to enhancing cybersecurity by incorporating obligations to cooperate among parties on the agenda.[103] However, as previously mentioned, cybersecurity can also be presented as a ground to justify data localisation requirements, thereby adding to the regulatory burden for businesses and SMEs. Trade agreements also tend to preserve policy rooms for parties to act in pursuance of 'essential security interests'. For example, Article 23 of the WTO JSI Agreement on Electronic Commerce extends the security exception provisions applicable on trade in goods and services also to e-commerce transactions.

2.5 Opportunities for EU SMEs in the global digital economy

Internationalisation of SMEs – helping them to benefit from the single market as well as connect to the global economy – is a long-standing priority in the EU, as the Small Business Act in 2008 already shows[104]. The European Court of Auditors (ECA) noted in 2022 that during 2014-2020, the EU spent roughly EUR 850 million to support SME internationalisation through a range of networks (e.g. Enterprise Europe Network (EEN), Startup Europe), region-specific support (e.g. SME centre for China), market access support (e.g. Access2Markets run by the DG TRADE) and SME help desks[105]. Digitalisation is a key component of the Commission's current SME strategy[106], which is linked to

[100] USTR, *USTR Statement on WTO E-Commerce Negotiations*, (*2023*), https://ustr.gov/about-us/policy-offices/press-office/press-releases/2023/october/ustr-statement-wto-e-commerce-negotiations.
[101] See section 2.2.1 above.
[102] Consolidated text: Regulation (EU) 2024/2847 of the European Parliament and of the Council of 23 October 2024 on horizontal cybersecurity requirements for products with digital elements and amending Regulations (EU) No 168/2013 and (EU) 2019/1020 and Directive (EU) 2020/1828 (Cyber Resilience Act), OJ L 2847, (2024), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02024R2847-20241120.
[103] CPTPP, Art. 14.16; RCEP Art. 12.13.
[104] European Commission, Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of the Regions – "Think Small First" – A "Small Business Act" for Europe, COM(2008) 394 final, (2008), https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A52008DC0394.
[105] European Court of Auditors, *SME Internationalisation Instruments: A Large Number of Support Actions but Not Fully Coherent or Coordinated,* (2022), pp. 10–14, https://data.europa.eu/doi/10.2865/036149.
[106] European Commission, Communication from the Commission to the European Parliament, The Council, The European Economic and Social Committee and the Committee of the Regions: An SME Strategy for a Sustainable and Digital Europe, COM (2020) 103 final, (2020), https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=COM%3A2020%3A103%3AFIN.

support measures through the EEN, strengthening SMEs access to the European Digital Innovation Hubs (EDIHs)[107] and financing of upskilling through the Digital Europe Programme[108]. A plethora of policies, programs and projects have been adopted at the Member State (MS) level to help SMEs connect to global market, many of which are dedicated to improving digital skills and support regarding digital transformations. The latest study on EU SMEs internationalisation strategy collected information on 251 relevant MS-level policy instruments, 104 of which focus on one or more specific industrial ecosystem[109]. Digital industry is one of the most targeted (68 %) ecosystems[110]. Given the volume and variety of efforts, building interconnection and coherence across the support measures is currently a key challenge[111].

In the context of digital transformation, support to SME internationalisation has to respond to multiple policy drivers. The changing nature of digital innovations and added geopolitical tensions mean that support policies need to be flexible and adaptable to changing circumstances over short periods of time. Indeed, digitalisation can improve SMEs business operations as well as address many of traditional barriers to internationalisation (e.g. lack of access, resources, information etc.). However, for traditional SMEs, digitalisation measures such as creating online presence, offering products and services for sale and providing consumer support can be skills and resource intensive – thereby requiring special attention. Currently, financial support comprises the majority (57 %) of all policy instruments supporting SME internationalisation. Beyond that advisory services (56 %), market introduction (51 %) and information services (51 %) are most offered[112]. Digital SMEs on the other hand may have specific support needs (e.g. skills training, access to appropriate technologies and solutions) that are not covered by financial or advisory supports – calling for more targeted assistance.

At the EU level, funds such as the European Regional Development Fund (ERDG) and Recovery and Resilience Fund (RRF) can also support SMEs digital transition. EU-level instruments help SMEs to participate in networks and consortia, enabling them to address specific challenges including digital transition. There are also initiatives specific to priority markets (e.g. USA, China, India) such as Trade and Technology Councils (TTC) for the USA and India. For example, since its inception in 2021 the EU-US TTC has established a dedicated working group on promoting SMEs' access to digital tools. In 2024, in the context of the 6th Ministerial meeting of the EU-US TTC, the working group issued a set of recommendations in line with its mandate. Issues covered by the recommendations include (i) enhancing digital-related trainings for SMEs, (ii) establishing transatlantic exchange

[107] European Commission, European Digital Innovation Hubs Network, https://european-digital-innovation-hubs.ec.europa.eu/get-know-us.
[108] European Commission, *The Digital Europe Programme*, https://digital-strategy.ec.europa.eu/en/activities/digital-programme.
[109] European Commission, *Study of the European Union SMEs Internationalisation Strategy: Final Report,* (2025), pp. 18–19, https://data.europa.eu/doi/10.2873/9687786 (hereinafter, 'EU SMEs Internationalisation Strategy').
[110] EU SMEs Internationalisation Strategy, (n 109) above, pp. 21–22.
[111] EU SMEs Internationalisation Strategy, (n 109) above, pp. 88–91; European Court of Auditors, (n 105) above, p. 38.
[112] EU SMEs Internationalisation Strategy, (n 109) above, pp. 22–23. As there are more than one policy approaches per instrument, when all the percentage values are added together, the total figure exceeds 100.

programmes, (iii) information sharing on cybersecurity, intellectual property and standards, and (iv) expanding access to financing opportunities for SMEs[113]. Connection to the US market, the Silicon Valley in particular, is highly significant in promoting SMEs access to digital technologies. The European Institute of Innovation and Technology (EIT) Hub Silicon Valley connects SMEs with the innovation ecosystem of California[114]. The EU SME Centre in China informs SMEs about the range of restrictions to be faced when accessing the Chinese market through e-commerce. The Intellectual Property (IP) Helpdesk advises and prepares SMEs that face challenges regarding IP issues in the Chinese market.

As mentioned above, the EU-level policies are further complemented by MS level market and industry-specific support policies that focus on digitalisation in different ways. One common attribute of such programs is that they are focused exclusively on the domestic firms. For example, the Study on SME Internationalisation Strategy identified 23 instruments divided among 12 EU Member States providing support for SME internationalisation to the US market. Sweden, maintaining one of largest number of MS level policies supporting SME internationalisation, also has the most (six altogether) US-market-specific programs. Many of these programs help SMEs connect to the technology firms in the US. Given that digital SMEs challenges and needs often vary, the Swedish approach has been to select the highest performing SMEs and providing them specifically tailored support to address challenges faced by them[115]. Other Member States also focus on the digital industrialisation of the SMEs. Portugal's program on Internationalisation via E-Commerce[116], and Spain's XPande Digital program[117] is noteworthy.

One of the key traditional roles of state commitments in trade agreements is to secure preferential access to the market/s of the other party(ies). In contrast, due to the generally open and borderless nature of the data flows underpinning the global digital economy, state commitments on digital trade such as ensuring free data flow, or prohibition of customs duties on electronic transmissions, are geared to preserve an already open system. Hence, unlike traditional trade, commitments on digital trade do not have the effect of increasing SMEs' (or other businesses) market access. Traditional market access promotional measures for SMEs such as information on available FTA preferences by the access2markets project of the DG TRADE, have no comparable twin in the digital domain yet.

Existing research on the traditional forms of trade suggests that SMEs may benefit from the greater market access opportunities and lower trade costs resulting from preferential agreements. However, easy access to the relevant information and the complexity of rules are the key challenges

[113] *Recommendations from the U.S.-EU Trade & Technology Council Working Group on Promoting SME Access to and Use of Digital Tools*, (2024) https://digital-strategy.ec.europa.eu/en/library/recommendations-us-eu-trade-technology-council-working-group-promoting-sme-access-and-use-digital.
[114] EIT Hub Silicon Valley, https://go-eit.eu/eit-silicon-valley-hub/
[115] EU SMEs Internationalisation Strategy, (n 109) above, p. 54.
[116] Portugal Digital, *Internationalization via e-commerce,* https://portugaldigital.gov.pt/en/accelerating-digital-transition-in-portugal/corporate-incentive-systems/internationalization-via-e-commerce/.
[117] Interreg Europe, *Xpande Digital Program,* (2018), https://www.interregeurope.eu/good-practices/xpande-digital-program.

in realising those benefits[118]. It remains unclear whether the same benefits and drawbacks will be reproduced int the effect of DTAs on SMEs. The commitments on non-imposition of customs duty on electronic transmissions will undoubtedly benefit all businesses, including SMEs. In contrast, EU's reluctance[119] to include commitments on non-discrimination of digital products can prove costly for SMEs seeking to internationalise. Apart from these commitments, existing preferential agreements provide for cooperation, exchange of information and greater access to open government data to enable SMEs to benefit from digital trade. Similar provisions are also contained in the WTO JSI Agreement on Electronic Commerce. It remains to be seen how much cooperation may take place between parties, which will actually benefit SMEs. More specific and SME-needs oriented mechanisms like the trade and technology councils, EEN and EDIH may prove beneficial. Experience gathered from such cooperation will provide useful examples on how to improve SME-specific support elements in trade agreements. Member States' experience in running various forms of incentive programs for digital SMEs will also be valuable feedback for developing more effective SME support commitments in preferential agreements.

3 Conclusions and recommendations

The landscape of digital economic transformation is complex and multi-layered, to which digital trade policy forms a significant and integral part. Within the EU, digital transformation has multiple dimensions – safety, security, protection of individual rights, fairness of access and competition and economic growth – triggering a plethora of regulatory responses which are driven by different objectives and priorities. EU digital regulation like the GDPR currently set the global standard for the protection of the individual in the cyberspace. For SMEs, the new digital regulatory environment can bring great benefits in terms of fair access to global markets and lowering of trade costs. In contrast, increased burden of regulatory compliance may deter new entrepreneurs from entering the digital economy. To tackle this, modern digital trade policy needs to be ambitious and integrative on one hand, and easily adaptable to changing circumstances on the other. Following are key recommendations that the European Parliament may wish to consider itself, or may call upon the Commission where appropriate, to improve the benefit of the digital economy for the SMEs.

Adopt a whole-of-government approach to digital regulation

- Ensure coherence between various regulatory frameworks (GDPR, AI regulation, Data Strategy) to prevent excessive compliance burdens on SMEs.

[118] Thibo Clicteur and others, *EU Trade Policy: How Can FTAs Better Deliver for SMEs?*, Briefing PE653.627, Policy Department for External Relations, European Parliament, (2021), pp. 11–13, https://www.europarl.europa.eu/thinktank/en/document/EXPO_BRI(2021)653627.

[119] There is no clear explanation yet as to why non-discrimination is absent in EU's core priorities on digital trade, or relevant commitments in DTAs or FTAs. One author points to a difference of views. Hannes Berggren, *Economic Security and Digital Trade: Time for an EU-US Digital Agreement,* Kommerskollegium, (2024), pp. 14,15, 31, https://www.kommerskollegium.se/globalassets/publikationer/rapporter/2024/economic-security-and-digital-trade.pdf. Another possible reason can be some EU member states' likely interest to protect certain domestic sectors that may fall under the coverage of 'digital product' (e.g. audiovisual products).

- Explore how to enhance the coordination of trade policy across various internal regulatory tools and policies. For example, for the single market, the Parliament may consider examining the pathways to ensure coherence between EU market access commitments (e.g. in trade in services) and the implementation of digital market regulation. With regard to external trade relations, the Parliament may consider exploring how the EU regulatory standards for digital markets can be internationalised through FTAs and DTAs.

Enhance SME-specific digital market access support

- Encourage the Commission to develop dedicated mechanisms to provide systematic support for SMEs in navigating digital trade regulations (similar to the existing 'access2market' mechanism). The Parliament may also consider recommending expansion of the mandates of SME Helpdesk and IP Helpdesk to assistance on digital market-related issues.
- Examining the extent to which current tiered responsibility and specific exemptions under the DSA has been effective in reducing compliance burden for the SMEs, would be a useful step. In addition, harmonising and streamlining the member-state-level requirements for compliance with the digital standards like the GDPR will lower costs for SMEs.
- Improve access to digital trade facilitation tools, including guidance on cross-border data flows and e-commerce platforms.

Continue EU leadership at the WTO on digital trade

- Continue efforts to make the moratorium on e-commerce duties permanent to prevent additional costs for SMEs engaged in digital trade.
- EU and other allies should engage with the persistent objectors on e-commerce issues at the WTO (e.g. India and South Africa) to find a pathway that secures the consensus required for the inclusion of the JSI Agreement on Electronic Commerce to the WTO rulebook.

Strengthen SME-specific provisions in EU FTAs and DTAs

- Provide explicit policy space for incentivising SME growth through innovation, ensuring fair competition with larger digital firms. One example could be creation of regulatory space (e.g. through exception clause) that would allow SMEs more flexibility (e.g. longer time periods) to comply with digital market standards or provide targeted digital trade facilitation support.
- Strengthen commitments for free cross-border data flows while balancing privacy and security concerns.
- Consider whether incorporation of non-discrimination commitments regarding digital products in preferential agreements is beneficial for SMEs.
- Embed SME-specific institutional cooperation, information and experience sharing commitments in FTAs and DTAs

Increase market access and global integration for SMEs

- Expand SME-focused trade and technology councils to provide tailored digital trade access support.
- Encourage Member States to share the best practices on promoting and enabling SMEs access to digital innovation ecosystems.

- Harmonise EU internal market rules (e.g. licensing etc.) so that an SME established in one member state market can seamlessly expand using the digital single market, which would strengthen it for global competition.

Promoting and enforcing fair competition in digital markets

- Enforce Digital Markets Act (DMA) provisions to prevent anti-competitive behaviours that hinder SMEs from fairly operating in digital platforms.
- Ensure major platforms provide equal visibility and access opportunities for SMEs offering apps, services and digital content.

Finally, the Parliament may also consider commissioning more detailed studies on the specific transformative aspects of trade resulting from digitalisation and their impact on SMEs.

Bibliography

Agreements, cases and laws

Agreement on Digital Trade Between the European Union and the Republic of Korea (EU-Korea DTA) (2025)

Agreement on Digital Trade Between the European Union and the Republic of Singapore (EU-Singapore DTA)

Case T-55/24, *Application – Meta Platforms Ireland v Commission,* Court of Justice of the European Union (CURIA), (2024)

Case T-58/24, *Application – Tiktok Technology v Commission*, Court of Justice of the European Union (CURIA), (2024)

Chapter 14: Electronic Commerce, Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP)' (2018)

Chapter 19: Digital Trade, Agreement between the United States of America, the United Mexican States, and Canada (USMCA) (2020)

Free Trade Agreement between the European Union and New Zealand (EU-New Zealand FTA), OJ L, 2024/866, (2024)

Marrakesh Agreement Establishing the World Trade Organization 1994 (1867 UNTS 154; 33 ILM 1144 (1994))

Regional Comprehensive Economic Partnership Agreement (RCEP) (2020)

Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC (Digital Services Act), OJ L 277, (2022)

Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector and amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act), OJ L 265, (2022)

Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation), OJ L 119, (2016)

Trade and Cooperation Agreement between the European Union and the European Atomic Energy Community, of the One Part, and the United Kingdom of Great Britain and Northern Ireland, of the Other Part (EU-UK TCA) OJ L 149 (2021)

Translation: Cybersecurity Law of the People's Republic of China (Effective June 1, 2017)

Academic sources

Bradford A., *The Brussels Effect: How the European Union Rules the World,* Oxford University press, (2020)

——, *Digital Empires: The Global Battle to Regulate Technology,* Oxford University Press, (2023)

Burri M., 'Digital Trade: New Design Elements in Preferential Trade Agreements', *The Concept Design of a Twenty-First Century Preferential Trade Agreement: Trends and Future Innovations,* Cambridge University Press, (2025)

Burri M., Vásquez Callo-Müller M. and Kugler K., 'The Evolution of Digital Trade Law: Insights from TAPED', *World Trade Review,* vol. 23, (2024)

Eifert M., Metzger A., Schweitzer H. and Wagner G., 'Taming the Giants: The DMA/DSA Package', *Common Market Law Review,* vol. 58, (2021)

Hashimzade N, Myles G and Black J, 'Contestable Market', *A Dictionary of Economics,* Oxford University Press, (2017)

EU documents and reports

Annual Report on European SMEs 2023/2024: SME Performance Review 2023/2024, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs (2024)

Briefing: The EU's Digital Trade Policy, EPRS, European Parliament, (2024)

Communication from the Commission to the European Parliament, The Council, The European Economic and Social Committee and the Committee of the Regions: An SME Strategy for a Sustainable and Digital Europe, COM (2020) 103 final, European Commission, (2020)

Communication from the Commission to the European Parliament, The Council, the European Economic and Social Committee and the Committee of the Regions – Trade Policy Review: An Open, Sustainable and Assertive Trade Policy 2021, COM(2021) 66 final, European Commission, (2021)

Commission Staff Working Document: Annual Single Market Report 2021, SWD(2021) 351 final, European Commission, (2021)

EU Trade Policy: How Can FTAs Better Deliver for SMEs?, EPRS, Policy Department for External Relations, (2021)

European Court of Auditors, *SME Internationalisation Instruments: A Large Number of Support Actions but Not Fully Coherent or Coordinated,* (2022)

European Investment Bank., *Investment Barriers in the European Union 2023: A Report by the European Investment Bank Group,* (2024)

Study of the European Union SMEs Internationalisation Strategy: Final Report, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, European Commission, (2025)

Study on Mapping Data Flows: Final Report, Directorate-General for Communications Networks, Content and Technology, European Commission, (2021)

Other documents and reports

Berggren H., *Economic Security and Digital Trade: Time for an EU-US Digital Agreement,* Kommerskollegium, (2024)

Brannon V.C. and Holmes E.N., *Section 230: An Overview*, Congressional Research Service, (2024)

Cheng W. and Brandi C., *Governing Digital Trade – a New Role for the WTO*, German Development Institute, (2019)

Latham & Watkins, China's New AI Regulations, (2023)

López-González J. and Jouanjean M-A., Digital Trade: Developing a Framework for Analysis, *OECD Trade Policy Papers*, vol. 205, (2017)

Monteiro J-A. and Teh R., Provisions on Electronic Commerce in Regional Trade Agreements, ERSD-2017-11, WTO, (2017)

OECD, *Digital Trade Inventory: Rules, Standards and Principles*, (2021)

——, *OECD Digital Economy Outlook 2024 (Volume 1): Embracing the Technology Frontier*, (2024)

Recommendations from the U.S.-EU Trade & Technology Council Working Group on Promoting SME Access to and Use of Digital Tools, (2024)

United Nations, Digital and Sustainable Trade Facilitation: Global Report 2023, (2023)

WTO, Ministerial Declaration on Trade in Information Technology Products, WT/MIN(96)/16, (1996)

——, Work Programme on Electronic Commerce, WT/L/274, (1998)

——, Ministerial Declaration on the Expansion of Trade in Information Technology Products, WT/MIN(15)/25, (2015)

——, Joint Statement Initiative on Electronic Commerce – Agreement on Electronic Commerce, INF/ECOM/87, (2024)

——, Work Programme on Electronic Commerce: Ministerial Decision, WT/MIN(24)/38; WT/L/1193, (2024)

Webpages

Commission Sends Preliminary Findings to X for Breach of DSA, European Commission

Digital Trade in EU Trade Agreements, European Commission

Digital Transformation, OECD

DMA Designated Gatekeepers, European Commission

European Data Strategy, European Commission

European Digital Innovation Hubs Network, European Commission

Germany: Berlin Court Orders Social Media Platform X to Share Public Data on Elections with Researchers, Library of Congress

The Digital Europe Programme, European Commission

USTR Statement on WTO E-Commerce Negotiations, United States Trade Representative

DIRECTORATE-GENERAL FOR EXTERNAL POLICIES
EXTERNAL POLICIES ANALYSIS AND SUPPORT

European Parliament

BRIEFING

How to strengthen innovation and intellectual property: Examples of Swedish SMEs and unicorns

ABSTRACT

Sweden has emerged as a hub for 'unicorn' startups, with companies like Klarna, Neko Health and Epidemic Sound leading the way in fintech, health tech and music tech. This case study explores the factors behind Sweden's success in fostering high-growth startups, including the availability of venture capital, a well-educated workforce, strong government support and close collaboration between academia and businesses. Sweden's digital infrastructure and its culture of innovation have enabled startups to thrive, with Stockholm playing a pivotal role in connecting entrepreneurs, investors and policymakers. By examining key success stories and analysing enabling factors, this study provides insights into how Sweden's 'unicorn' ecosystem contributes to the global digital economy. Lessons drawn from Sweden's approach can inform strategies to strengthen innovation and intellectual property in the European Union, enhancing competitiveness in the fourth global industrial revolution.

EP/EXPO/2023/OP/0001/INTA/LOT6/1/C/4-2

EN

June 2025 – PE 754.479

© European Union, 2025

External Policies Analysis and Support Unit, Directorate-General for External Policies

AUTHORS

- Peter-Tobias STOLL, Georg-August University, Göttingen
- Zaker AHMAD, Georg-August University, Göttingen

RESEARCH ASSISTANCE

- Jan THOMSEN, Georg-August University, Göttingen

PROJECT COORDINATOR (CONTRACTOR)

- Universität Innsbruck (PARLATRADE Consortium)

This paper was requested by the European Parliament's Committee on International Trade (INTA).

The content of this document is the sole responsibility of the authors, and any opinions expressed herein do not necessarily represent the official position of the European Parliament.

CONTACTS IN THE EUROPEAN PARLIAMENT

Coordination: Rasma KASKINA, External Policies Analysis and Support Unit

Editorial assistant: Balázs REISS

To give feedback or obtain copies, please write to exas@europarl.europa.eu.

VERSION

English-language manuscript completed in June 2025.

COPYRIGHT

Brussels © European Union, 2025

Reproduction and translation for non-commercial purposes are authorised, provided the source is acknowledged and the European Parliament is given prior notice and sent a copy.

© Image on cover page used under licence from Adobe Stock.

This paper will be published on the European Parliament's online database, 'Think Tank'.

Table of contents

1 Introduction 4
2 Swedish ‘unicorns’ at a glance 4
3 Success stories 6
4 Enabling factors 8
5 Lessons learnt and conclusions 12
6 Recommendations 13
References 15
Annex 16

1 Introduction

'Unicorns' are start-up companies with a market value of more than one billion US dollars. Swedish 'unicorn' startups have gained international recognition for their contributions to the digital economy. Sweden's ability to produce high-value startups stems from a unique combination of economic policies, a strong innovation environment and a robust intellectual property framework. The country's success can be attributed to various factors, including its highly skilled workforce, availability of venture capital (VC) and government initiatives that promote entrepreneurship. Additionally, the culture of collaboration between startups, academia and investors has played a critical role in fostering growth. Notably, Swedish consumers are early adopters of innovation, further supporting startup expansion. This case study examines key Swedish 'unicorns', including Klarna and Neko Health, to illustrate how their business strategies and enabling conditions have driven success. By identifying lessons from Sweden's startup ecosystem, this paper aims to provide recommendations for strengthening innovation and intellectual property frameworks across the European Union.

2 Swedish 'unicorns' at a glance

Since 2011, following 12 companies (Table 1) have acquired the title of 'unicorn' in Sweden[1]. Although the sample is too small to establish any statistically significant trend, most (nine out of 12) of the Swedish 'unicorns' can be observed to be based in Stockholm. Although, London is by far the leading European city in producing 'unicorns', Stockholm is currently ranked fourth, placed right behind Paris and Berlin[2]. Table 1 below also shows that the 2020-2022 period has produced most (eight out of 12) of the existing Swedish 'unicorns'. This trend is well-aligned with the Europe-wide flow of investment into startups, which is on a declining trend since hitting a peak in 2022[3]. Another shared feature is the technology (tech) intensive nature of all the startups, even though their specialisations vary to a significant extent.

Table 1. List of Swedish startups achieving 'unicorn' status

Company	Domain	'Unicorn' date	City	Sector	Value (billion USD)
Neko Health	nekohealth.com	Jan. 2025	Stockholm	HealthTech, Healthcare IT	1.80
Polarium	polarium.com	May 2022	Stockholm	Energy Tech, Energy Storage Tech	1.00
Instabox	instabox.io	Apr. 2022	Bromma	Consumer, Logistics Tech	1.08

[1] For additional details, see Annex, Table A.

[2] Deep Ecosystems, *Startup Heatmap Europe,* 2025, p. 15, https://pub.marq.com/268897e7-8d94-45e1-ba7f-9d1a02c93e3c/ #2d7255Y0ODdk.

[3] Deep Ecosystems, (n 2) above, p. 16.

Company	Domain	'Unicorn' date	City	Sector	Value (billion USD)
Einride	einride.tech	Jan. 2022	Stockholm	Energy Tech, Electric Vehicles	1.40
Voi	voi.com	Dec. 2021	Stockholm	Energy Tech, Electric Vehicles	1.00
Neo4j	neo4j.com	Jun. 2021	San Mateo, Malmo	Enterprise Infrastructure, Big Data Infrastructure	2.00
Kry	kry.health	Apr. 2021	Stockholm	HealthTech, Healthcare Platforms	2.00
Epidemic Sound	epidemicsound.com	Mar. 2021	Stockholm	Consumer, Music Tech	1.40
Oatly	oatly.com	Jul. 2020	Malmo	Food and Agriculture Tech, Food Tech	2.00
Northvolt	northvolt.com	Jun. 2019	Stockholm	Energy Tech, Energy Storage Tech	11.75
Klarna	klarna.com	Mar. 2014	Stockholm	FinTech, Alternative Lending	14.60
Spotify	spotify.com	Jun. 2011	Stockholm	Consumer, Music Tech	8.53

Source: *Tracxn Technologies (2025).* ***Note:*** *Grey highlighted companies have graduated from the 'unicorn' status. See Annex 1 for more information.*

Some other companies, such as Spotify and Oatly, have earlier been 'unicorns'. But they have exited this category, as these companies are now publicly traded. Acquisition is another way to graduate from the 'unicorn' status. Skype, established in 2003, was sold to Microsoft in 2011 for USD 8.5 billion. Another recent 'unicorn', Northvolt, has been partially acquired by Scania in April 2025, after its bankruptcy[4].

Table 2. Leading countries by number of 'unicorns' (2024)[5]

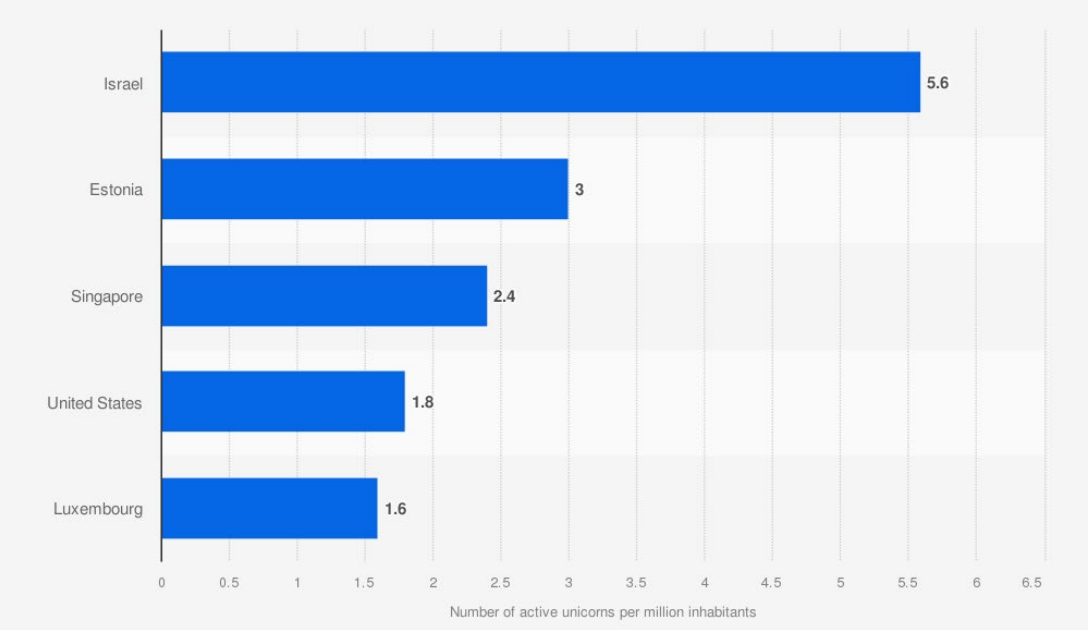
Country	Number of 'unicorns' as of 2024
United States	656
China	168
India	71
United Kingdom	53

[4] 'Volkswagen's Scania Buys Bankrupt Northvolt's Heavy Industry Battery Packs Unit' *Reuters* (11 April 2025), https://www.reuters.com/business/autos-transportation/volkswagens-scania-buys-bankrupt-northvolts-heavy-industry-battery-packs-unit-2025-04-11/.

[5] The table was created using data from https://www.statista.com/statistics/1096928/number-of-global-unicorns-by-country/.

Country	Number of 'unicorns' as of 2024
Germany	30
France	26
Israel	25
Canada	21
Brazil	17
Singapore	16
South Korea	14
Australia	9
Switzerland	9
Netherlands	9
Mexico	8
Ireland	7
Indonesia	7
Japan	7
Sweden	6
Hong Kong	6

Figure 4. Countries with the highest number of startup 'unicorns' per million inhabitants as of 2023[6]



3 Success stories

In order to ascertain the enabling factors that lead to the emergence of 'unicorns', it is useful to consider individual case examples from Sweden. For that purpose, we look at two Swedish 'unicorns'. One is Klarna – the longest-standing as well as the most valuable Swedish 'unicorn' at

[6] The graph was created using data from https://www.statista.com/statistics/1608305/countries-highest-number-starup-unicorns-world-per-capita/.

present. The other is Neko Health – the youngest 'unicorn' in the sable, achieving its status in early 2025.

3.1 Klarna

Klarna is an innovative fintech company founded in 2005, as a 'buy now, pay later' (BNPL) service provider. It also offers solutions to merchants, enabling the latter to offer flexible payment options to customers. Over the past decade, Klarna attracted investments from more than 130 investors, including VC firms (e.g. Sequoia Capital, Blackrock) and individuals, reaching a valuation of USD 45.6 billion in 2021, which was followed by a sharp decline. The company started to make profits again in 2024, achieving a turnover of USD 2.81 billion and a profit of USD 21 million. An initial public offer (IPO) of shares in the US is currently under preparation.

Klarna founders have been closely affiliated to the Stockholm School of Economics (SSE). Their business plan was originally developed in the SSE Business Lab[7]. The company is known for its customer-centric approach and user-friendly design based on a highly interconnected technology ecosystem. Within the EU, the expansion of Klarna has taken place gradually, informed by factors such as market size and consumer demand and implemented through partnerships with e-commerce platforms as well as acquisition of similar local service providers. The financial activities of Klarna are conducted under different licenses granted by EU member states and remain under supervision of the relevant bodies[8]. The company is well-aware of the compliance costs imposed by EU regulations such as the general data protection regulation (GDPR) and the EU consumer credit directive (CCD), as well as the unforeseeable challenges arising from misunderstanding and misinterpretation of those rules[9]. In 2024, the Swedish Financial Supervisory Authority (*Finanzinspektionen*) issued an approximate USD 46 million fine for not complying with anti-money laundering regulation[10]. Klarna has also expressed concerns over fragmented regulation of artificial intelligence (AI) in the EU. It called for "harmonised, consistent, quick and clear decisions under EU data regulations that enable European data to be used in AI training for the benefit of Europeans"[11].

[7] SSE Business Lab, https://www.hhs.se/en/outreach/sse-initiatives/sse-business-lab.
[8] Klarna, Governance Oversight, https://investors.klarna.com/governance/Governance-Oversight/default.aspx#:~:text=Klarna%20Bank%20AB%20is%20authorized,and%20by%20establishing%20local%20branches.
[9] Klarna Holding AB: Prospectus regarding the admission to trading of SEK 500,000,000 Floating Rate Tier 2 Notes, (2023), pp. 9-12, https://www.fi.se/sv/vara-register/prospektregistret/GetFile/?id=23-15360&format=application/pdf&name=23-15360.pdf#:~:text=This%20Prospectus%20has%20been%20approved%20and%20registered,the%20prospectus%20to%20be%20published%20when%20sec.
[10] Finanzinspektionen, Klarna receives a remark and an administrative fine, (11 December 2024), https://www.fi.se/en/published/sanctions/financial-firms/2024/klarna-receives-a-remark-and-an-administrative-fine/.
[11] Klarna, A Call for AI Innovation: Why Fragmented Regulations Are Holding Europe Back, (19 September 2024), https://www.klarna.com/international/press/a-call-for-ai-innovation-why-fragmented-regulations-are-holding-europe-back/.

3.2 Neko Health

Neko Health is a health technology company founded in 2018 and later incorporated in Sweden (2023) and the United Kingdom (UK) (2024). Through its health centres currently located only in Stockholm and London, the company offers 'preventative healthcare' – a quick, yet comprehensive full body scan that can facilitate preventative diagnosis of many health conditions such as diabetes or skin cancer[12]. Its innovative, AI-assisted, non-invasive full-body scanning and assessment solution was publicly launched in 2023. It is planned that the company will expand into more health centres across Europe.

The company has been founded by Spotify CEO Daniel Ek. It demonstrates that founders and investors can play a crucial role in bringing about new and innovative startups in an interconnected environment. Since its incorporation, Neko Health has engaged in two rounds of funding, raising a total of USD 325 million[13]. It has been invested by several VC firms led by Lightspeed. The company's current valuation stands at USD 1.8 billion.

4 Enabling factors

There are a number of descriptive and analytical studies as well as few scientific publications that examine the factors contributing to the success of Swedish 'unicorns'. Before delving deeper into these factors, some caveats must be mentioned. While studies identify socio-economic factors closely associated with the success of the Swedish tech 'unicorns', the main contribution of those factors lie in collectively producing an environment conducive to innovation. Any of these factors should not be considered in isolation as 'causing' the success of 'unicorns'. Moreover, success or failure of these policy combinations in promoting 'unicorns' may depend on additional contextual factors (e.g. size of the economy, or the scale of public spending). Keeping these in mind, following are the key attributes that can be identified as having a positive impact on the success of the Swedish 'unicorns'.

4.1 Company level and business strategies

Often, the relevance of the company or product strategy is highlighted. In this context, flexibility, high quality, profit orientation instead of growth orientation, flat hierarchies with a strong informal network structure and a strong customer orientation are often mentioned[14].

[12] Neko, Neko Health Launches in London, (4 September 2024), https://www.nekohealth.com/se/en/neko-health-launches-in-london.
[13] Neko, Neko Health raises $260m Series B, https://www.nekohealth.com/se/en/neko-health-raises-260m-series-b.
[14] Anna Hammarberg, The Land of Unicorns, Business Sweden, (9 January 2020), https://www.business-sweden.com/insights/ articles/the-land-of-unicorns/.

4.2 Resources and infrastructures: Capital, human resources, infrastructure and knowledge

The availability of new, innovative ideas as well as highly qualified personnel to bring those to fruition, are of foundational importance for the emergence of any tech startup. Sweden excels on both these aspects. The country currently ranks 2nd in the Global Innovation Index (GII)[15]. In the GII sub-categories of infrastructure[16] and business sophistication[17], Sweden ranks 1st worldwide. According to the European Patent Office (EPO) data, Sweden is one of the top-10 origins of digital technology patent filings, with particular strength in the digital communication sector[18]. In addition, Swedish educational institutions perform significantly above the OECD average. The ease of recruiting staff from abroad and a making welcoming environment available for them are also important. In 2025, Swedish policy to attract high-skilled workers in line with the EU Blue Card policy has been further relaxed[19]. It should also be mentioned in this context that social security is well developed in Sweden, which positively contributes to the willingness to work on risky projects.

Once a startup gains traction, its exponential scale-up in financial terms to become a 'unicorn' is heavily dependent on the ability to access VC. This is one of the main reasons behind the European 'unicorns' emergence in key capitals and financial hubs such as London, Paris and Berlin. This may also partially explain the tendency of the Swedish 'unicorns' to expand into these regions. However, much of the investment in Swedish 'unicorns' come from VC firms in the United States (US), which are attracted to Sweden's successful startup activity. The latter was enabled by early efforts from the City of Stockholm to create business incubators as well as support from the European Investment Bank[20]. Furthermore, investors, their expertise, their positive attitude towards digital economy projects and their networking with each other and with companies and entrepreneurs play a key role.

Next to capital and human resources, the technical infrastructure is important, including internet connectivity and energy and related services. Even more important is the availability of knowledge and technology. Both these elements can hardly be seen only as separated input factors. They more likely can be ascribed as an ecosystem, where knowledge and technologies is kept by but also

[15] WIPO, Global Innovation Index 2024, p. 230, https://www.wipo.int/web-publications/global-innovation-index-2024/assets/67729/2000%20Global%20Innovation%20Index%202024_WEB3lite.pdf.
[16] This includes ICT infrastructure (access and use), general infrastructure and ecological sustainability. See ibid.
[17] This includes, among others, knowledge-intensive employment, public research-industry collaboration, level of high-tech and ICT services imports. See ibid.
[18] See Annex, Table B.
[19] Among others, the salary threshold has been reduced to 1.25 times than 1.5 times of average Swedish salary. The period of employment has also been reduced to 6 months from 1 year. See Migrationsverket, New EU Blue Card rules now apply, (7 January 2025), https://www.migrationsverket.se/nyheter/news-archive/2025-01-07-new-eu-blue-card-rules-now-apply.html#:~:text= The%20salary%20threshold%20is%20reduced,a%20new%20EU%20Blue%20Card.
[20] Emma Björner, Olof Zetterberg, *Stockholm: Tale of the Unicorn Factory,* European Investment Bank, (2019), pp. 14-15, https://www.eib.org/files/publications/country/city_transformed_stockholm_en.pdf.

widely developed and shared within and among the companies but also in exchange with academic institutions.

4.3 The innovation ecosystem: A role for academia and consumers

The close interaction of companies with academic institutions in an enabling environment goes beyond maintaining a knowledge base but also involves support of the development of startups by such institutions. Close and mostly informal interaction help to bring about and further develop innovation and related startups[21]. As is often maintained, the Stockholm area is a particular active spot, where entrepreneurs, academics and companies – but also investors are close and can interact easily[22]. This non-hierarchical entrepreneurial environment is further enhanced by access to digital technologies, which simplifies the collaboration process and opens opportunities for new solutions. This has been termed as a 'digital entrepreneurial ecosystem' (DEE)[23].

On the demand side, a DEE benefits from the existence of a thriving consumer base that is interested to and capable of adopting new technology-based solutions. The Swedish domestic consumer base, which is highly educated and enjoys one of the highest levels of network connectivity worldwide, works an asset for the local 'unicorns'. Early adoption of cutting edge technologies also add to this demand-pull. For example, the Swedish government's early adoption of digital banking, electronic identification and online payment mechanism facilitated the rise of a thriving private fintech market[24]. The pull of the big and digitally well-serviced markets may also explain the Swedish 'unicorns' prioritisation of expansion to big EU member states. Lastly, the openness of the Swedish society and the global culture of its workforce and consumers also facilitates innovations that are born ready for international markets.

4.4 Facilitation by government measures

Both national and sub-national governments play pivotal roles in facilitating the entrepreneurial ecosystem in Sweden. The Swedish government has supported this development in many ways. It has promoted the digitalisation of society and the economy by expanding infrastructure, through targeted measures for education and training[25], but also by promoting the availability of hardware in households and businesses (e.g. subsidisation of personal computer purchase)[26].

[21] Sweden ranked 2nd of the 133 economies in the Global Innovation Index (GII) of the WIPO, see https://www.wipo.int/gii-ranking/en/sweden.
[22] For instance, Stockholm School of Economics (SSE) top 2 in Europe for creating Unicorn founders, see https://www.hhs.se/en/about-us/news/sse-business-lab/2023/sse-tech-founders/.
[23] Ana Venancio, Winnie Picoto, Inês Pinto, 'Time-to-unicorn and digital entrepreneurial ecosystems', *Technological Forecasting & Social Change,* vol. 190, (2023), pp. 2-4.
[24] Ultiro AB, *The Swedish Market: Financial Technology,* National Board of Trade (2024), p. 12, https://www.kommerskollegium.se/globalassets/publikationer/market-studies/market-study-financial-technology.pdf.
[25] Stockholm: Tale of the Unicorn Factory, (n 2020) above, pp. 7-13.
[26] Colm Fulton and Supantha Mukherjee, *Focus: How Sweden became the Silicon Valley of Europe,* Reuters, (11 August 2021), https://www.reuters.com/business/finance/how-sweden-became-silicon-valley-europe-2021-08-11/.

In addition, the state has specifically promoted startups on a large scale. The Swedish government actively fosters a thriving startup ecosystem through a range of strategic initiatives and funding programs. Key among these is Vinnova, Sweden's innovation agency, which offers substantial grants — ranging from SEK 500 000 to SEK 20 million — for startups focused on cutting-edge technologies, sustainability and digital transformation. The Swedish Energy Agency also supports green innovation with funding up to SEK 10 million for energy-efficient solutions. In the life sciences sector, SciLifeLab Boost provides SEK 90 million annually to biotech startups and researchers[27]. Almi Invest, Sweden's largest public venture capital firm, plays a crucial role by investing in scalable, high-growth startups with funding between EUR 90 000 and EUR 1 million[28]. The government also supports Clusters of Excellence, which bring together academia and industry to drive innovation, with SEK 1.2 billion allocated annually through 2028[29]. Complementing these efforts are equity-free grants, low-interest loans, tax incentives for research and development (R&D) and government-backed incubators and accelerators.

It should also be recalled that Sweden is one of the EU member states with the largest number of policy measures supporting SME industrialisation. As discussed in the briefing 'EU's Trade and Digital Economy – Challenges and Opportunities for Small and Medium Enterprises (SMEs)', Sweden identifies high-growth potential SMEs and provides them with tailored support for expansion. Such support mechanisms can be beneficial for 'unicorn' startups at the earliest stages of development. One such program, 'Scale US' helps SMEs already established their presence in the US market to scale up their operation[30].

The Swedish domestic taxation regime has both advantages and downsides for the operation of 'unicorns'. High income tax may make it difficult for startups to attract talents from overseas[31], However, the government uses the tax-revenue to provide various forms of infrastructural (e.g. high-speed network coverage) and social supports (e.g. childcare), improving the quality-of-life and social security for the startup stakeholders. Moreover, the government offers various forms of tax incentives[32], which include tax deductions to promote R&D, expert tax relief, lower tax rates on stock options earned by startup staff[33]. The government also provides venture capital and export credit support to its startups. Current rate of corporate tax in Sweden is 20.6 %, which is lower than

[27] See, e.g., https://www.strata.team/main-swedish-grants-to-boost-technology-in-2025/.
[28] Almi-Invest, https://www.almi.se/en/venture-capital/.
[29] Government offices of Sweden, Excellent research and innovation are encouraged in largest-ever research and innovation bill, (16 December 2024), https://www.government.se/press-releases/2024/12/excellent-research-and-innovation-are-encouraged-in-largest-ever-research-and-innovation-bill/.
[30] Business Sweden, Scale US, https://www.business-sweden.com/industries/consumer-goods/scale-us/.
[31] The Reuters report (n 26 above) mentions Spotify considering moving out of Sweden for this reason.
[32] Business Sweden, Incentives Guide, (2024), pp. 6-7, https://www.business-sweden.com/globalassets/services/learning-center/establishment-guides/2024/guides-and-other-regulations/incentives-guide.pdf#page=6.24.
[33] Backer McKenzie, 2022 updates to Sweden's tax treatment of stock options for startup employees and board members — qualified employee stock option, Lexology, (2022), https://www.lexology.com/library/detail.aspx?g=5c449742-f81f-4647-a456-5ef6a8eff875.

other European 'unicorn' hubs like United Kingdom (25.0 %), Germany (29.9 %), France (25.8 %) and Netherlands (25.8 %)[34].

The Swedish government actively engages in dialogues to promote tech startups, as evidenced by its convening of a conference in 2023, on policy measures to enable the growth of deep-tech start-ups[35]. The objective of the conference on deep-tech entrepreneurship was to provide a forum for knowledge exchange within the framework of the European innovation agenda and the action plan on intellectual property. The conference brought together private-sector stakeholders (deep-tech start-ups, corporates, venture capitalists), national and regional governments and EU institutions and stakeholders to discuss the challenges and possible solutions[36]. The conference report called upon the European Commission and coming Council presidencies to take systematic and pan-European action that bring together and align stakeholders to improve Europe's deep tech ecosystem[37].

5 Lessons learnt and conclusions

These various findings and insights shed a light on the success of Swedish 'unicorns' and its underlying enabling factors. From the business perspective, while all Swedish 'unicorns' are by definition of high financial value and provide technology-oriented innovative solutions, their expertise vary. Therefore, in technological terms, each unicorn's pathway to success may be unique.

The success of Swedish 'unicorns' can hardly be explained by one specific cause. It is highly likely that such success results from a number of factors, and even more so from their combination. Several lessons can be drawn from these insights.

An enabling entrepreneurial environment creates the conditions for tech startups to emerge. Stable, long-term national and sub-national policies that focus on creating sound physical and network infrastructure, affordable and high-quality education, social security and establishing linkages between public research and private market are very important. These policies create the framework conditions in which individuals and firms feel able to take entrepreneurial risks. The beneficial impact of such enabling environment would spread economywide and not be limited only to 'unicorns'. It should also be noted that in case of Sweden, successful creation of the enabling entrepreneurial environment is made possible by its high tax earnings and welfare-oriented public spending.

To become a 'unicorn', a successful startup has to scale up exponentially over a short period of time by attracting VC investments. Side-by-side, it has to strategically expand operations to new

[34] Christina Enache, Corporate Income Tax Rates in Europe, 2025, Tax Foundation Europe, (January 14 2025), https://taxfoundation.org/data/all/eu/corporate-income-tax-rates-europe/#:~:text=On%20average%2C%20the%20European %20countries,was%2023.5%20percent%20in%202024.

[35] Josefin van der Meer, Kjell-Håkan Närfelt, *Accelerating European deep tech*, Vinnova – Sweden's innovation agency, (2023), https://www.vinnova.se/globalassets/publikationer/vino-0039_rapport_deeptech-slutleverans-231009-tillganglighetsanpassad.pdf?cb=20231009153522.

[36] See European Commission, Digital Decade Country Report 2024: Sweden, S. 16.

[37] Van der Meer and Närfelt, (n 35) above, p. 34-35.

markets with attractive consumer base and similar regulatory environment. Guided by these two factors, the 'unicorns' are likely to prioritise attracting US VC firms, while entering markets in the US, the UK, Germany, France and the Netherlands. From a trade policy perspective, here is a lesson to focus on easing trade in financial services and promoting market access in digital services with major trade partners such as the US and the UK. Moreover, commitments such as ensuring free cross-border data flow, removal of duties on e-commerce transaction, ensuring interoperability and paperless trade, which are currently pursued as core EU digital trade policy priorities[38], would also be beneficial for the 'unicorns'.

Regarding expansion into EU member states' markets, a 'unicorn' is likely to take a dual approach – complying with the EU regulations (e.g. GDPR), on one hand, and addressing state-specific regulatory alignment (e.g. registration, licensing etc.), on the other. As seen in the example of Klarna, the 'unicorns' are well-aware of the costs arising from fragmented regulation. However, given their capacity, we assume the costs are relatively less prohibitive for 'unicorns' compared to an SME.

As the 'unicorns' provide innovative products and services, effective protection of their intellectual assets in the EU and abroad is key. As statistics of the international patent applications show[39], Sweden has a leading position in securing its digital technology through patents. However, such activities are led by more mature, publicly traded corporations (e.g. Ericsson, Volvo etc.). Also, companies going through the 'unicorn' phase are highly unlikely to enter markets where intellectual property protection is not ensured.

6 Recommendations

The European Parliament may consider encouraging the EU-wide adoption of such practices to promote innovative startups and SMEs, the success of which have been proven by the growth of Swedish 'unicorns'. In this connection, the Parliament may choose to explore how the lessons from the Swedish experience can be translated into policy packages. Instead of nationwide reforms, such policies could even consider targeting its top universities and public research institutions, and build a startup environment around it. Political initiatives have already been taken, which seek to harmonise the best policy practices to promote startups across the members states[40]. Such efforts deserve broader endorsement and support.

To ensure that EU digital trade policy continues to enable and facilitate the growth of European 'unicorns', the Parliament may

- continue ensuring that key digital trade policy priorities (e.g. free data flow, removal of customs duties) are embedded and implemented in trade agreements with major economies. In this regard, deepening digital trade ties with the UK, and maintaining formal

[38] Marc Jütten, *The Eu's digital trade policy,* EPRS, European Parliament, (2024), pp. 8-10 https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/757615/EPRS_BRI(2024)757615_EN.pdf.
[39] See Annex, Table B
[40] See, for example, Startup Nations Standard, https://startupnationsstandard.eu/about.html; Also see, European Startup Nations Alliance, https://esnalliance.eu/.

and informal collaboration with the digital ecosystem stakeholders (e.g. entrepreneurs, VC firms, individuals) in the US could be prioritised.

- explore the pathways for integrating interoperability of digital standards (e.g. payment standards) as a commitment in trade agreements.
- examine further and ensure that foreign providers of key financial services (e.g. venture capitals) have reasonable and free access to member state markets. In this connection, existence of a domestic enabling environment and successful startup activities will be helpful to attract foreign VC firms.

We draw attention to the recommendations made in the briefing 'EU's Trade and Digital Economy – Challenges and Opportunities for Small and Medium Enterprises (SMEs)', which are also beneficial for 'unicorns'. Effective enforcement of EU digital regulation by member states to ensure fair access to digital markets for all and promoting a digital single market will help the 'unicorns' by lowering regulatory fragmentation. Recommendations related to SME internationalisation, support and improving market access will benefit early-stage startups, which are the potential 'unicorns' of the future.

Next to digital trade policy, the Swedish best practice involves facilitative measures in policy areas such as migration and taxes, which the EP might wish to consider accordingly. One possible area could be to examine whether there is room for better or more flexible implementation of the EU Blue Card Directive to enable member states to access the global pool of high-skilled workers.[41] The tax and spending patterns in Sweden may also deserve closer study for its replicability in other European contexts and impact on innovation.

Further studies should be undertaken to fully understand, what the EU at large can learn from the stunning success of Swedish unicorns. Among others, further sector or industry-specific comparison of Swedish regulatory environments with other EU member state frameworks could be undertaken.

[41] This could prove even more consequential in the face of potential exodus of talented workforce from the United States, in view of recent political events taking place there.

References

Björner E. and Zetterberg O., *Stockholm: Tale of the Unicorn Factory*, European Investment Bank, (2019)

____, *Swedish startups have doubled in value in the last five years, with impact taking the center stage*, Press Release, (22 February 2024)

____, *Incentives Guide*, (2024)

Deep Ecosystems, *Startup Heatmap Europe,* (2025)

European Commission, *Digital Decade Country Report 2024: Sweden*, (2024)

Hammarberg, A., *The Land of Unicorns, Business Sweden*, (9 January 2020)

Hodgson-Coyle, N. *Exploring the World of Swedish Unicorns and Soonicorns*, Blog, (03 July 2023)

Klarna Holding AB, *Prospectus regarding the admission to trading of SEK 500,000,000 Floating Rate Tier 2 Notes,* (2023)

van der Meer, J., Närfelt, K-H. *Accelerating European deep tech*, Vinnova – Swedens' innovation agency, (2023)

Skog, A., Lewan, M., Karlström, M., Morgulis-Yakushev, S., Lu, X., Teigland, R. *Chasing the Tale of the Unicorn – A study of Sweden's misty meadows*, Center for Strategy and Competitiveness, Stockholm School of Economics, Institute for Research Stockholm, (2016)

Ultiro AB, *The Swedish Market: Financial Technology*, National Board of Trade (2024)

Venancio A., Picoto W. and Pinto I., 'Time-to-unicorn and digital entrepreneurial ecosystems', *Technological Forecasting & Social Change,* vol. 190, (2023)

WIPO, *Intellectual property statistical country profile 2023: Sweden*, (2024)

Annex

Table A. Swedish startups acquiring 'unicorn' status (2011-2025)

No.	Name	Domain name	Overview	Unicorn date	City	Sector	Company stage*	Funded	Total funding (mil. USD)	Latest valuation (bil. USD (date))
1	Neko Health	nekohealth.com	Platform offering solutions for preventive health scans	Jan 23, 2025	Stockholm	HealthTech, Healthcare IT	Series B	Yes	325.37	1.80 (Jan 23, 2025)
2	Polarium	polarium.com	Provider of lithium-based hybrid energy storage solutions	May 3, 2022	Stockholm	Energy Tech, Energy Storage Tech	Series D	Yes	273.56	1.00 (May 3, 2022)
3	Instabox	instabox.io	Provider of e-commerce delivery services	Apr 21, 2022	Bromma	Consumer, Logistics Tech	Series C	Yes	304.71	1.08 (Apr 22, 2022)
4	Einride	einride.tech	Manufacturer of autonomous electric trucks	Jan 8, 2022	Stockholm	Energy Tech, Electric Vehicles	Series C	Yes	652.30	1.40 (Jan 8, 2022)
5	Voi	voi.com	App-based platform offering electric kick-scooter rental services	Dec 20, 2021	Stockholm	Energy Tech, Electric Vehicles	Series D	Yes	544.27	1.00 (Dec 21, 2022)
6	Neo4j	neo4j.com	Cloud based graph database platform	Jun 17, 2021	San Mateo, Malmo	Enterprise Infrastructure, Big Data Infrastructure	Series F	Yes	581.10	2.00 (Jun 17, 2021)
7	Kry	kry.health	Online platform offering video appointments with doctors and psychologists for mental health care	Apr 27, 2021	Stockholm	HealthTech, Healthcare Platforms	Series D	Yes	703.49	2.00 (Apr 27, 2021)
8	Epidemic Sound	epidemicsound.com	Cloud-based music licensing platform	Mar 11, 2021	Stockholm	Consumer, Music Tech	Series D	Yes	476.00	1.40 (Mar 11, 2021)
9	Oatly	oatly.com	Provider of plant-based dairy alternatives from oats	Jul 14, 2020	Malmo	Food and Agriculture Tech, Food Tech	Public	Yes	232.46	2.00 (Jul 14, 2020)
10	Northvolt	northvolt.com	Manufacturer of lithium-ion batteries for electric vehicles	Jun 1, 2019	Stockholm	Energy Tech, Energy Storage Tech	Acquired	Yes	9 000.00	11.75 (Jun 9, 2021)
11	Klarna	klarna.com	Developer of an online platform for point-of-sale financing	Mar 14, 2014	Stockholm	FinTech, Alternative Lending	Series H	Yes	4 190.00	14.60 (Oct 28, 2024)

No.	Name	Domain name	Overview	Unicorn date	City	Sector	Company stage*	Funded	Total funding (mil. USD)	Latest valuation (bil. USD (date))
12	Spotify	spotify.com	On-demand music streaming platform	Jun 17, 2011	Stockholm	Consumer, Music Tech	Public	Yes	2 063.00	8.53 (Jun 10, 2015)
	*	To raise capital, a startup company can go through several series of funding rounds (Series A, B, C ...). Eventually the company may offer its shares to the public, known as 'initial public offer' (IPO). Otherwise, it could be acquired by another company. In both cases, the startup graduates from the unicorn status (highlighted grey above).								

Source: *Tracxn Technologies (2025)*

Table B. Top 10 digital technology patent applicants in 2024 (by country)

Computer technology

Rank	Country		2024	2023	Change
1	US	United States	5 776	5 185	11 %
2	CN	P.R. China	2 395	2 565	-7 %
3	DE	Germany	1 558	1 383	13 %
4	KR	R. Korea	1 469	1 471	0 %
5	JP	Japan	1 307	1 412	-7 %
6	FR	France	786	899	-13 %
7	GB	United Kingdom	579	515	12 %
8	CH	Switzerland	463	337	37 %
9	NL	Netherlands	390	372	5 %
10	SE	Sweden	296	311	-5 %

Electrical machinery, apparatus, energy

Rank	Country		2024	2023	Change
1	KR	R. Korea	2 506	2 165	16 %
2	CN	P.R. China	2 454	1 856	32 %
3	JP	Japan	2 077	1 916	8 %
4	DE	Germany	2 031	2 136	-5 %
5	US	United States	1 769	1 809	-2 %
6	CH	Switzerland	813	670	21 %
7	FR	France	693	719	-4 %
8	NL	Netherlands	654	686	-5 %
9	GB	United Kingdom	332	289	15 %
10	SE	Sweden	279	229	22 %

Digital communication

Rank	Country		2024	2023	Change
1	US	United States	4 628	4 859	-5 %
2	CN	P.R. China	4 387	4 695	-7 %
3	KR	R. Korea	1 582	1 498	6 %
4	JP	Japan	1 205	1 208	0 %
5	SE	Sweden	1 025	1 456	-30 %
6	FI	Finland	813	816	0 %
7	DE	Germany	621	641	-3 %
8	FR	France	346	347	0 %
9	NL	Netherlands	250	261	-4 %
10	CH	Switzerland	204	166	23 %

Source: *EPO, Patent Index 2024, https://www.epo.org/en/about-us/statistics/data-download.*

PE 754.479
EXPO/2023/OP/0001/INTA/LOT6/1/C/4

Print ISBN 978-92-848-2872-2 | doi: 10.2861/5334378 | QA-01-25-140-EN-C
PDF ISBN 978-92-848-2871-5 | doi: 10.2861/2282904 | QA-01-25-140-EN-N